



Two-Dimensional Incompressible Flow in an Asymmetric Channel with Inclined Expansion and Contraction: A Parametric Study of Grid Sensitivity and Reynolds Scaling

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Abstract

This report presents a two-dimensional Computational Fluid Dynamics (CFD) study of steady, laminar, incompressible flow inside an asymmetric channel with inclined expansion and contraction backward-facing step using OpenFOAM v14. A nine-case parametric study was conducted to evaluate grid sensitivity and scaling effects across three Reynolds numbers ($Re = 60, 80, \text{ and } 100$). Spatial features were identified via boundary skin friction zero-crossings ($\tau_{w, x} = 0$) across three structured mesh levels ranging from 3,933 to 61,560 hexahedral cells. The variation in total wall footprint length between the medium and fine grids was approximately 3.4%, indicating that the key flow parameters are relatively insensitive to further mesh refinement over the tested range. Parametric velocity scaling showed that the total wall footprint length expanded from 1.81 to 4.57 units as the Reynolds number increased. Concurrently, the reattachment length measured from Point A reaches a maximum value of 5.07 units, while the vortex-core height shifted upward from approximately 0.38 to 0.74 units. The global continuity error dropped to a negligible level ($\approx -9.24 \times 10^{-16}$), showing mass conservation to discrete numerical limits.

1. Introduction

1.1 Theoretical Background

The study of fluid flow through an asymmetric channel with inclined expansion and contraction serves as a robust benchmark configuration for validating computational fluid dynamics (CFD) codes and tracking complex shear layer mechanics. Unlike a conventional backward-facing step with an abrupt 90-degree drop, this sloped transition geometry allows the boundary layer to detach naturally along the incline face under the influence of an adverse pressure gradient ($\frac{\partial p}{\partial x} > 0$), forcing the fluid momentum near the boundary wall to drop rapidly.



Because the localized inertial fluid forces can no longer overcome the wall viscous resistance, the boundary layer detaches from the solid boundary at a distinct separation point. The detached fluid generates a high-shear layer that curls backward, creating a primary recirculation zone (a localized vortex structure) characterized by negative stream velocities ($u < 0$) near the wall surface. Downstream of this recirculation bubble, the expanding fluid stream eventually recovers momentum and reattaches flush back onto the channel wall floor at a definitive reattachment point, transitioning back to standard forward channel flow. Tracking the localized structure of this recirculation zone, specifically mapping where separation begins, where reattachment occurs, and where the primary vortex eye forms are essential for capturing internal energy losses and viscous shear structures in expanding channels.

1.2 Geometric Parameter Identification

To initialize the simulation domain, the spatial constraints of the expansion channel were mapped directly from the discrete coordinates provided in the baseline Part 1 problem description dictionary (*blockMeshDict*). The domain is mathematically bounded along the horizontal direction by $X \in [0, 36]$ and vertically by $Y \in [0, 5]$. By tracing the individual coordinate entries mapped within the foundational topology vertices array, the complete spatial profile of the channel was systematically mapped into two interconnected flow regimes: an upstream entrance block and a downstream expansion channel.

The geometry structure is explicitly mapped into five hexahedral blocks defined across twenty-four unique vertex index nodes:

- **Upstream Channel Domain (X = 0 to X = 10):** The entry channel runs horizontally for a length of 10.0 units. It is vertically constrained between $Y = 2.0$ and $Y = 5.0$, defining a strict inlet patch gate height of exactly 3.0 units.
- **Asymmetric Incline Step Face (X = 10 to X = 13):** Between the horizontal coordinates of 10.0 and 13.0, the lower bounding wall forms a linear, inclined expansion drop. The wall surface slants downward from the coordinates (10, 2) to (13, 0), lowering the baseline height of the floor boundary from $Y = 2.0$ down to $Y = 0.0$. This geometric change implements a total vertical step drop height of exactly 2.0 units.
- **Downstream Expansion Channel (X = 13 to X = 36):** Following the step base corner at $X = 13.0$, the expanded fluid domain extends downstream for an additional straight run length of 23.0 units. The vertical boundaries span from the bottom floor at $Y = 0.0$ up to the ceiling wall at $Y = 5.0$, resolving a total post-expansion height profile of exactly 5.0 units.

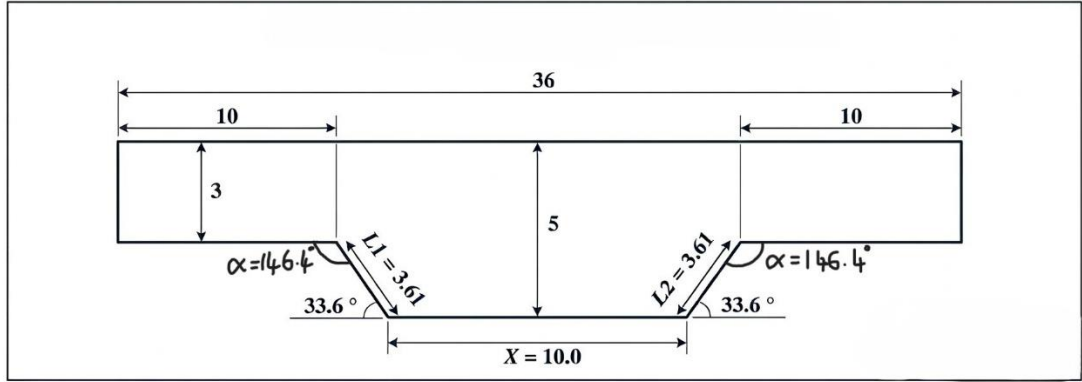


Fig 1.1: Geometric coordinate schematic of the asymmetric channel with inclined expansion and contraction boundaries.

The exact computational envelope layout and coordinate boundaries are visually detailed in the schematic framework above. The flow enters from the left through the inlet patch height $H_1 = 3.0$ units and passes through the flat upstream channel until it hits the expansion start vector at Point A ($X_A = 10.0$, $Y = 2.0$). The lower boundary then slants downward along the inclined expansion wall, dropping past the localized boundary layer Separation Point until it reaches the base step corner at Point B ($X_B = 13.0$, $Y = 0.0$). This expands the vertical height profile of the central main domain to a maximum channel profile of $H_2 = 5.0$ units. Within this expanded central boundary zone, the detached shear layer forms the primary recirculation vortex before expanding back flush onto the horizontal floor at the Reattachment Point. The fluid core then continues downstream past the contraction step running from $X = 23.0$ to 26.0 before discharging through the final exit channel core outlet boundary at $X = 36.0$.

1.3 Block Layout and Structural Mesh Decomposition

To execute a stable finite-volume solution, the continuous 2D space was mapped and solved through multi-block topological structural decomposition. The total channel envelope was divided into five distinct hexahedral blocks, hand-coded to ensure perfect cell-face alignment and strict mass conservation across the interfaces:

- Block 0 (Upstream Entry Core): Spans horizontally from $X = 0$ to 10 , vertically from $Y = 2$ to 5 , and extends across the out-of-plane direction from $Z = 0$ to 1 . The downstream end of this section, at $(X, Y) = (10, 2)$, corresponds to Point A defined in the problem statement and serves as the reference location for measuring the reattachment length. This section contains the upstream channel through which the prescribed inlet velocity enters the computational domain.
- Block 1 (Sloped Expansion Region): Extends from $X = 10$ to 13 . The upper wall remains horizontal at $Y = 5$, while the lower wall slopes downward from $(10, 2)$ to $(13, 0)$. Consequently, the channel height increases continuously from 3.0 to 5.0 units across this section. This inclined expansion generates the separated-flow region downstream of the expansion.

- Block 2 (Expanded Central Region): Extends from $X = 13$ to 23 , with the channel bounded by $Y = 0$ at the lower wall and $Y = 5$ at the upper wall. This is the main expanded section of the geometry in which the primary recirculation region develops and eventually reattaches to the lower wall.
- Block 3 (Sloped Contraction Region): Extends from $X = 23$ to 26 . The upper wall remains at $Y = 5$, while the lower wall slopes upward from $(23, 0)$ to $(26, 2)$. The channel height therefore decreases continuously from 5.0 to 3.0 units, returning the geometry toward its original channel height.
- Block 4 (Downstream Exit Core): Extends from $X = 26$ to 36 , with the channel height maintained at 3.0 units between $Y = 2$ and $Y = 5$. This downstream section provides sufficient distance for the flow to recover after the contraction before reaching the outlet boundary at $X = 36$.

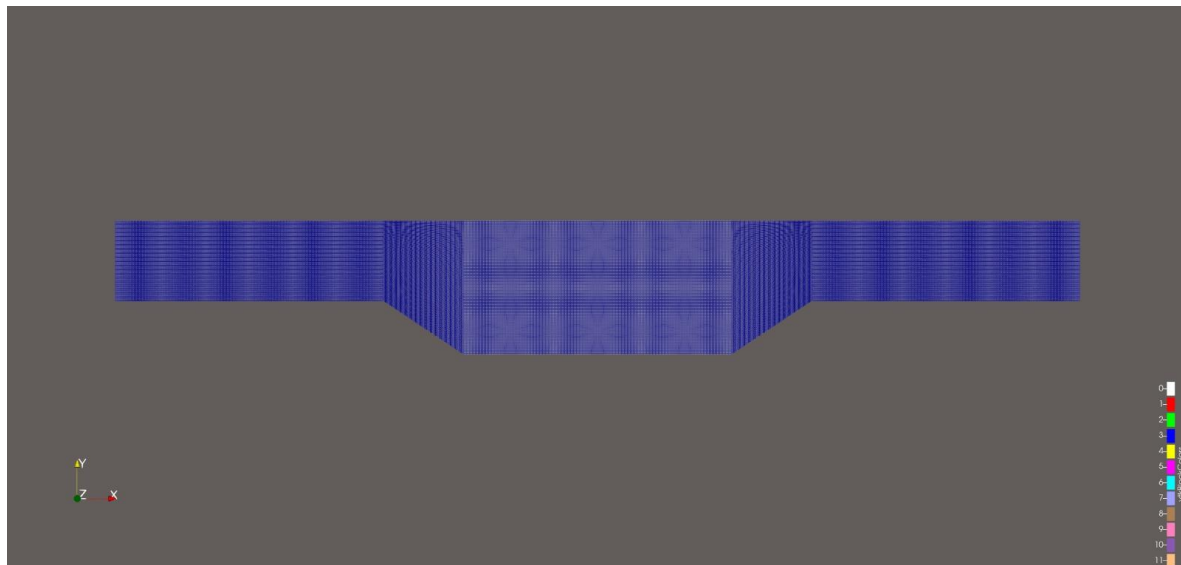


Fig 1.2: Mesh of $Re = 60$ with fine mesh

2. Mathematical Modelling and Governing Laws

2.1 Governing Equations

The steady-state, laminar, viscous flow field through the asymmetric expanding channel domain is modelled mathematically using the two-dimensional incompressible Navier-Stokes differential transport equations. Assuming a constant kinematic fluid viscosity and the complete absence of external body forces (such as gravitational or electromagnetic fields acting on the fluid masses), the continuum mechanics equations simplify to the conservation of mass and linear momentum transport systems, as described in standard fluid mechanics literature (Versteeg and Malalasekera, 2007)



The individual equations are formulated as separate, centred lines to preserve structural clarity:

Where u represents the horizontal velocity component along the direction of flow (m/s), v denotes the transverse velocity component normal to the channel walls (m/s), p indicates the localized static gauge pressure (N/m²), ρ is the uniform fluid density (kg/m³), and ν is the kinematic fluid viscosity (m²/s).

2.2 Wall Shear Stress Vector and Separation Metrics

To map the geometric structure of the primary recirculation zones and identify flow separation boundaries, the localized skin friction distributions are tracked across the solid channel patches. The wall shear stress vector is evaluated directly on the wall boundaries using OpenFOAM's native tensor projection utilities, rather than assuming simplified directional velocity derivatives. The tangential component of this stress vector along the local wall direction is tracked to identify separation and reattachment events through precise zero-crossings.

On the horizontal downstream channel floors (where the boundary unit normal vector is aligned strictly with the vertical direction, $\hat{n} = \hat{j}$), the streamwise component $\tau_{w, x}$ directly represents the relevant skin-friction direction:

$$\tau_{w,x} = \mu \left(\frac{\delta u}{\delta y} \right)_{y=y_{wall}}$$

The spatial progression of this tangential wall shear stress component provides the definitive indicators required to define the recirculation layout bounds:

- **Separation and Flow Detachment:** Where the adverse pressure gradient completely depletes forward fluid momentum along the sloped incline, the local tangential shear stress switches sign through a zero-crossing, initiating a reverse backflow regime inside the separation zone.
- **Flow Reattachment:** The precise coordinate downstream where the expanding shear layer re-contacts the flat horizontal wall boundary marks the end of the recirculation bubble. At this node, the streamwise boundary stress component $\tau_{w, x}$ undergoes a second zero-crossing, returning from negative values back into positive forward-flow parameters.

2.3 Spatial Geometry Length Formulations

Utilizing the localized zero-crossing indices extracted from the boundary data plots, the structural length metrics of the recirculation envelope are calculated using standard Euclidean distances:



- **Inclined Wall Contribution to the Recirculation Footprint (C_1):** Measures the linear, straight wall separation distance directly along the solid sloped surface, stretching from the initial separation coordinate (X_{sep} , Y_{sep}) down to the step baseline corner node at (13.0, 0.0):
- **Horizontal Floor Contribution to the Recirculation Footprint (C_2):** Tracks the horizontal distance from the base step corner ($X = 13.0$) out to the flat wall reattachment point ($X_{reattach}$):
- **Total Vortex Wall Footprint Length (C):** Combines both solid surface metrics to compute the cumulative boundary path length exposed to reverse backflow dynamics along the channel walls:
- **Core Vortex Vector Distance (L_1):** Measures the absolute distance from the primary vortex centre core eye coordinate (X_v , Y_v) to the geometric midpoint centre of the steep slope located at (11.5, 1.0):

3. Computational Methodology and Numerical Setup

3.1 Grid Discretization Framework

Spatial discretization of the five-block fluid domain was hand-coded using OpenFOAM's utility tool, *blockMesh*, executed via the standard utilities natively packaged within the open-source library environment (Greenshields et al., 2024). The mesh structure is composed entirely of pure structured hexahedral cells, ensuring formal spatial alignment with the channel axis and eliminating non-orthogonality errors within the shear layer core.

To systematically evaluate numerical convergence, a three-level grid density matrix was constructed using identical spatial boundaries while scaling cell distributions proportionally along the direction indices:

- **Coarse Mesh Resolution:** Initialized with a baseline block mesh layout containing a cell grid configuration of (45 23 1) and (18 23 1), resolving a total density profile of 3,933 cells.
- **Medium Mesh Resolution:** Scaled uniformly by a spatial tracking factor of two, defining block sizes of (90 45 1) and (36 45 1) to resolve a total grid distribution of 15,390 cells.



- **Fine Mesh Resolution:** Configured at maximum refinement to capture steep spatial boundary layers, utilizing cell counts of $(180\ 90\ 1)$ and $(72\ 90\ 1)$ to generate a total grid layout of 61,560 cells.

All three meshing passes utilized uniform mesh spacing throughout the channel volume (*simpleGrading (1 1 1)*), maintaining uniform, cell spacing within each block through *simpleGrading (1 1 1)* across the geometric expansion zone to avoid artificial numerical dispersion. Before solving, grid quality metrics were verified through the *checkMesh* utility, confirming zero topological errors, maximum cell openness limits under 1.60×10^{-16} , and a maximum mesh non-orthogonality bounded safely at 33.54° .

3.2 Boundary Conditions Specification Matrix

To match fluid characteristics with the continuous governing transport equations, specific initial and boundary conditions were defined within the OpenFOAM *0/* directory files for the velocity field vector (*0/U*) and the static pressure scalar (*0/p*).

The boundary properties are explicitly mapped as follows:

- **Inlet Patch Gate:** A uniform, fixed velocity profile vector field (*type fixedValue*) was applied directed entirely along the horizontal component. To evaluate Reynolds scaling metrics across the laminar flow cycle, the inlet velocity magnitude (U_{in}) was scaled linearly according to the standard scaling parameter $U_{in} = Re \cdot \nu / D_h$ (where D_h is the hydraulic diameter), yielding exact tracking parameters:
 - Reynolds Number 60 Run: $U_{in,60} = 1.6000 \times 10^{-4}$ m/s
 - Reynolds Number 80 Run: $U_{in,80} = 2.1333 \times 10^{-4}$ m/s
 - Reynolds Number 100 Run: $U_{in,100} = 2.6667 \times 10^{-4}$ m/s
 The accompanying static pressure property at the inlet patch was set to a zero-gradient boundary condition (*type zeroGradient*).
- **Outlet Patch Boundary:** Configured at the channel exit plane to enforce fully developed flow profiles. The static gauge pressure scalar field was locked to an absolute zero value reference field (*type fixedValue; value uniform 0*), paired with a zero-gradient boundary tracking condition applied to the velocity vector (*type zeroGradient*).
- **Walls Patch Envelope:** Enforces the standard physical boundary condition across all solid ceiling, floor, and steep incline faces. Velocity components were bound to an absolute standstill field (*type noSlip*), paired with a zero-gradient pressure condition (*type zeroGradient*).
- **FrontAndBack Patch:** Extruded across a single unit cell depth ($Z = 1.0$) and locked explicitly to an *empty* geometric type parameter. This boundary flag instructs the OpenFOAM solver engine to ignore all partial derivatives along the third axis ($\delta/\delta z = 0$), enforcing a true two-dimensional solution space.

3.3 Numerical Discretization Schemes and Solver Properties



The governing transport matrices were solved using the steady-state, incompressible finite-volume solver engine, *incompressibleFluid* (OpenFOAM v14). Velocity-pressure cross-coupling was handled using the standard Semi-Implicit Method for Pressure-Linked Equations (SIMPLE) numerical algorithm, utilising the foundational pressure-velocity coupling methodologies established by Patankar (1980). The convective momentum transport fluxes were resolved using the high-bounded, second order upwind bounded scheme, ensuring high matrix stability across the shear layer.

The finite volume discretization schemes inside the *system/fvSchemes* file were configured as follows:

- **Time Integration:** Set to steady-state tracking (*ddtSchemes: default steadyState*).
- **Gradient Transformations:** Discretized using second-order central differences (*gradSchemes: default Gauss linear*).
- **Divergence Operators:** Convective momentum transport fluxes were resolved using the native OpenFOAM syntax (*divSchemes: div (phi, U) Gauss bounded upwind*). The convective momentum term was discretised using a bounded upwind scheme, ensuring high matrix stability across shear layer.
- **Laplacian Operators:** Viscous diffusion terms were solved using standard linear Gaussian schemes with explicit non-orthogonal corrections (*laplacianSchemes: default Gauss linear corrected*).

Linear matrix solvers within *system/fvSolution* configured Geometric-Algebraic Multigrid solvers (*GAMG*) to handle pressure updates, paired with smooth solvers for velocity vector tracking. Numerical execution limits inside *system/controlDict* were bounded to a maximum allocation limit of 3,000 iterations for the dense fine meshes to avoid early termination. To ensure deep physical convergence, automatic simulation cutoff bounds (*residualControl*) were locked to an ultra-strict parameter threshold:

Pressure Field Residual Target (p) = 1.0×10^{-6}

Velocity Component Residual Target (U_x, U_y) = 1.0×10^{-7}

3.4 Solver Convergence and Continuity Verification

To guarantee that the extracted flow physics represent a mathematically valid solution to the continuous Navier-Stokes equations, the internal convergence behaviour was strictly monitored during runtime execution. The finite-volume solver algorithm maps two primary indicators to verify convergence: row residual decay profiles and local mass continuity conservation metrics.

The standalone mathematical metrics are explicitly detailed below:

3.4.1 Residual Target Compliance

The global cell residuals for the horizontal velocity components (U_x), vertical components (U_y), and the pressure field variable (p) demonstrated a smooth, monotonic decay across all case operations.



- For the coarse and medium grid layouts, the internal fields satisfied the automated residual control constraints within 333 to 1,000 iterations, causing the solver loop to execute a clean, premature termination.
- For the highly dense fine mesh cases (~61.5k cells), the enhanced local spatial gradients required an extended calculation timeline. These cases safely utilized the hard runtime ceiling of 3,000 iterations, allowing the initial linear solver errors to drop dramatically from baseline thresholds down to asymptotic parameter bounds:

Final U_x ; Initial Residual $\approx 1.09 \times 10^{-7}$ to Final Residual $\approx 7.89 \times 10^{-9}$

Final U_y ; Initial Residual $\approx 2.19 \times 10^{-7}$ to Final Residual $\approx 1.56 \times 10^{-8}$

Final p ; Initial Residual $\approx 1.22 \times 10^{-7}$ to Final Residual $\approx 3.80 \times 10^{-8}$

3.4.2 Mass Continuity Conservation Metrics

True physical convergence requires strict adherence to the incompressible continuity equation ($\nabla \cdot U = 0$). At the termination of the fine grid calculations, local flux imbalances across individual finite-volume cells were evaluated using the built-in time-step continuity error utility tracker. The global continuity imbalance was reduced to approximately -9.24×10^{-16} , indicating excellent discrete mass conservation across the multi-block channel configuration.

3.5 Post Processing and Extraction of Flow Characteristics

To systematically extract the structural features of the primary recirculation zone from the converged finite-volume fields, a standardized post-processing protocol was executed within the ParaView data analysis environment. OpenFOAM's native *wallShearStress* runtime function object was applied to the boundary patches to compute the local wall shear stress vector field directly on the solid boundary faces, rather than estimating values from the interior fluid cells. The specific engineering parameters required by the project framework were extracted using the following tracking protocols:

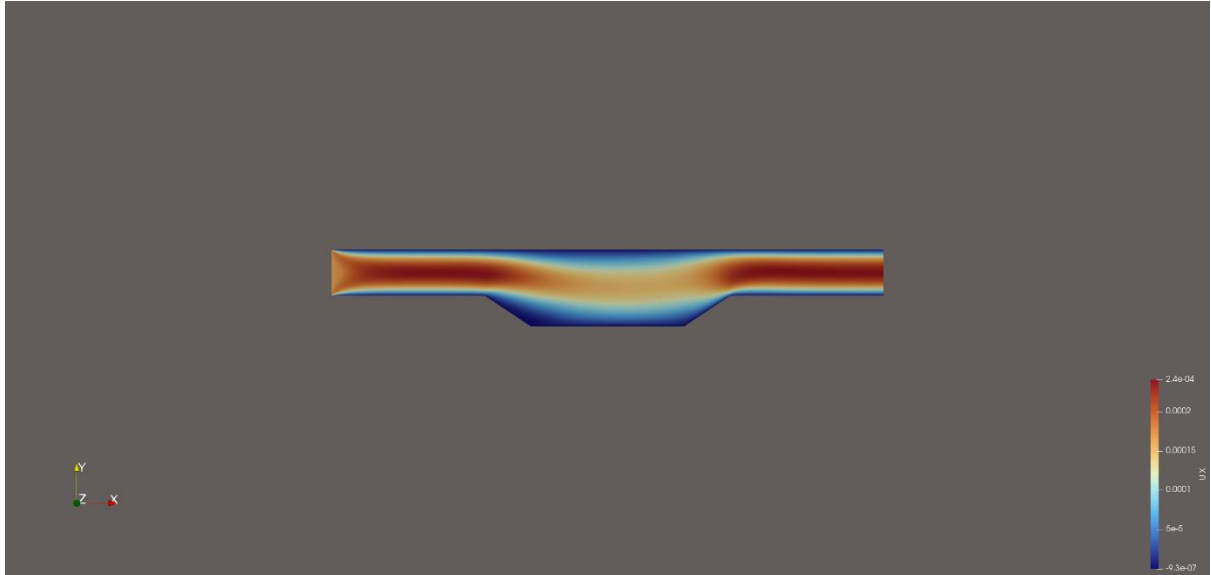


Fig 3.1: Contour of Velocity along X direction of $Re = 60$ with fine mesh

3.5.1 Flow Reattachment Length from Point A

The localized flow reattachment point (x_r) was isolated by plotting the streamwise wall shear stress component ($\tau_{w, x}$) directly along the downstream horizontal channel floor patch ($Y = 0$). The exact reattachment node was identified by locating the spatial zero-crossing intersection where the skin friction profile transitions from a negative to a positive value ($\tau_{w, x}$: negative to positive). Following identification, the reattachment length metrics from Point A ($L_{r, A}$) were calculated relative to the initial expansion drop boundary at $X = 10.0$ using the following expression:

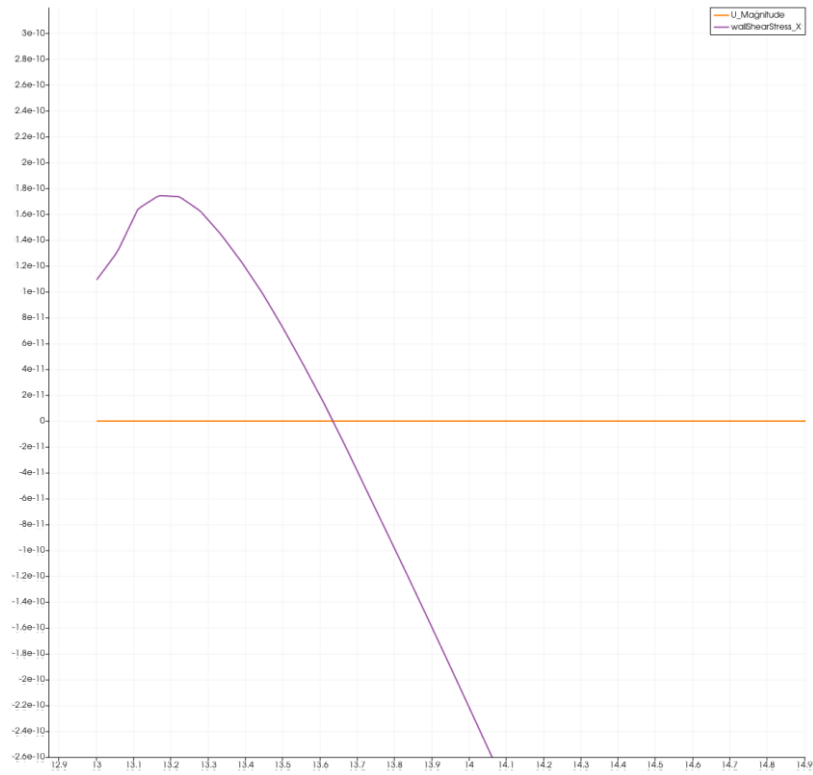


Fig 3.2: wallShearStress_x of $Re = 60$ with fine mesh along the horizontal floor from which reattachment length from Point A ($L_{r,A}$) and C_2 is calculated

3.5.2 Primary Vortex Size Relative to Channel Walls

To characterize the cumulative footprint of the recirculating flow zone, the total boundary footprint length (C) was calculated by separating the solid channel boundaries into two distinct surface contributions:

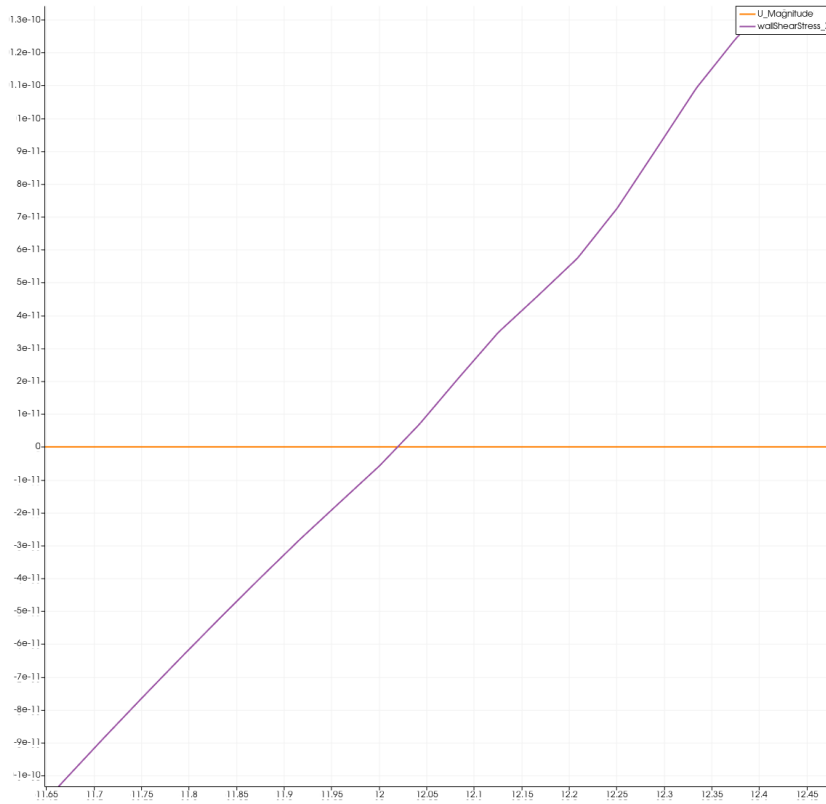


Fig 3.3: wallShearStress_x of $Re = 60$ with fine mesh along the slope from which C_1 is calculated

The individual values were extracted via the following metrics:

- **Inclined-Wall Contribution (C_1):** The initial flow separation node (x_s , y_s) was pinpointed by locating the corresponding zero-crossing where the local tangential component of the wall shear stress vector field transitions from positive to negative along the sloped face. Because the wall is inclined, (C_1) represents the straight-line distance along the sloped surface from the separation point to the bottom corner at Point B (13.0, 0.0), calculated via the Euclidean distance formula:
- **Horizontal Floor Contribution (C_2):** This parameter measures the linear horizontal distance running from the base step corner at Point B ($X = 13.0$) out to the downstream zero-crossing reattachment node (x_r):

3.5.3 Primary Vortex Centre and Distance from L_1

The primary vortex centre core coordinate pair (x_v , y_v) was isolated by applying a line-probe sweep across the recirculation zone to identify the spatial point corresponding to the absolute minimum streamwise velocity value within the reverse backflow domain ($u_{core} = u_{min} < 0$). To determine the core displacement

relative to the sloped geometry, the absolute distance (L_1) was calculated from this vortex centre (x_v, y_v) to the geometric midpoint centre of the inclined wall section located at (11.5, 1.0) using the distance formula:

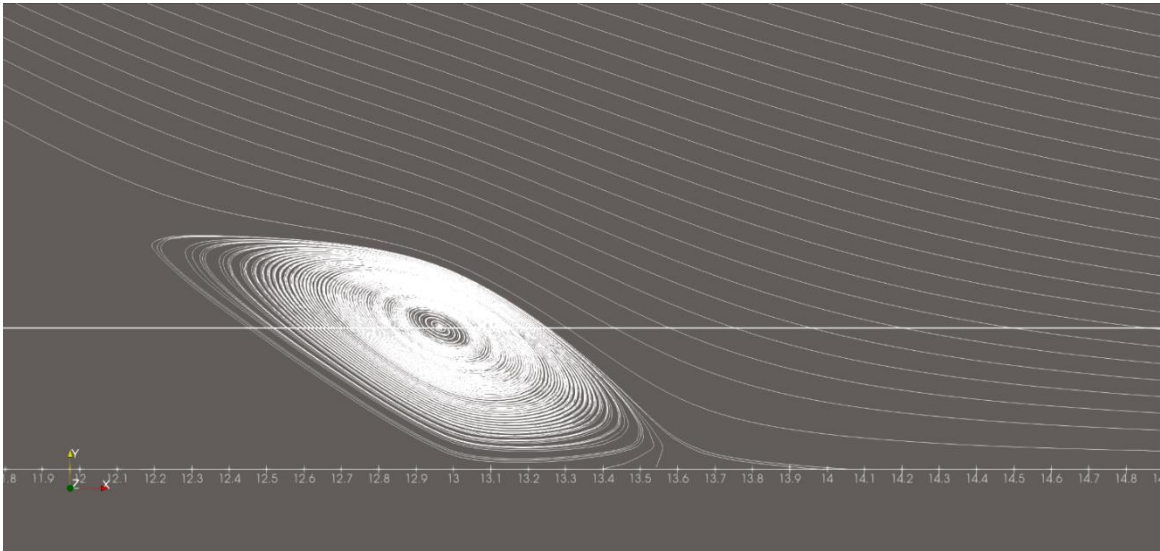


Fig 3.4: Vortex formation of $Re = 60$ with fine mesh depicted using Streamlines to approximate the centre of vortex visually

4. Results and Discussion

4.1 The Parametric Evaluation Grid

The numerical features extracted across all nine computational testing configurations are compiled in the master database matrix layout below. All parameters are evaluated at the final steady-state iteration limits for each case:

Table 1: Table containing the results of Parametric Evaluation of asymmetric channel with inclined expansion and contraction boundaries.

Reynolds Number	Grid Tier	Mesh	Total Cells	Hex	Reattachment Length from Point A (X = 10.0) in the direction of flow	Size of Primary Vortex relative to the Channel Walls (C)	Distance: Vortex Center to Slope Midpoint (L_1)
Re = 60	Coarse		3,933		3.570	1.75	1.55
	Medium		15,390		3.610	1.75	1.66
	Fine		61,560		3.630	1.81	1.68
Re = 80	Coarse		3,933		4.360	3.44	1.57

Re = 100	Medium	15,390	4.340	3.42	1.58
	Fine	61,560	4.335	3.125	1.59
	Coarse	3,933	5.140	4.68	1.69
	Medium	15,390	5.080	4.57	1.70
	Fine	61,560	5.070	4.57	1.70

4.2 Mesh Independence Analysis ($Re = 60$ as Baseline)

A grid convergence evaluation was conducted at $Re = 60$ to analyse the sensitivity of the results to cell density scaling. The reattachment length from Point A ($L_{r,A}$) tracked at 3.57 (coarse) and 3.61 (medium), settling at 3.63 units on the fine mesh layout. The total vortex wall footprint length (C) plateaued at 1.81 units.

The reattachment length changed from 3.61 to 3.63 units between the medium and fine meshes, corresponding to a variation of approximately 0.55%. The total vortex wall footprint changed from 1.75 to 1.81 units, corresponding to approximately 3.4%. These small variations indicate that the extracted flow characteristics are relatively insensitive to further mesh refinement over the tested range.

4.3 Verification Scope and Validation Limitation

The present study focuses on numerical solution verification through grid-independence analysis. Direct experimental validation was not performed because a matching reference geometry and Reynolds-number definition could not be established with sufficient confidence for the present configuration. Therefore, the reported results are assessed primarily through mesh-independence, solver convergence, and physical consistency of the computed flow features rather than through quantitative comparison with experimental measurements.

4.4 Parametric Evaluation of Reynolds Scaling

Following the grid sensitivity study on the fine mesh levels, flow characteristics were evaluated at higher Reynolds numbers using the step-graded velocity profiles ($U_{in,60}$ to $U_{in,80}$ to $U_{in,100}$). The data trends demonstrate a clear relationship between higher fluid momentum and the physical expansion of the primary recirculation zone:

- **Separation and Reattachment Trends:** As inertial forces increased, the flow separation node (x_s) shifted upstream along the sloped face from 12.02 down to 10.923. Concurrently, the downstream floor reattachment location past Point B ($L_{r,B}$) extended from 0.63 units to 2.07 units.
- **Recirculation Envelope Elongation:** Reflecting these shifts, the horizontal recirculation-zone extent (L_{recirc}) expanded from 1.61 units to 4.147 units, while the total vortex footprint length along the wall surface (C) grew from 1.81 units to a plateau of 4.57 units.



- **Vortex Core Translation:** The vortex-core location shifted upward as the Reynolds number increased, with the vertical coordinate (y_v) increasing from approximately 0.38 to 0.74 units, while the horizontal core position (x_v) showed minimal variance.

5. Conclusions and Outcomes

This computational fluid dynamics study successfully evaluated steady, laminar, incompressible flow across a two-dimensional asymmetric channel with inclined expansion and contraction using OpenFOAM v14. The engineering outcomes are summarized as follows:

- **Grid Sensitivity:** The three-level mesh density study demonstrated the asymptotic stabilization of the primary spatial flow metrics. At $Re = 60$, the total reattachment length from Point A stabilized at 3.63 units. The variation between the medium and fine grids for the total vortex wall footprint length (C) was limited to 3.4%, indicating that the key flow predictions are relatively insensitive to further mesh refinement.
- **Reynolds Scaling Dynamics:** Scaling the velocity across the operational parameters ($Re = 60, 80, 100$) showed a direct relationship between rising fluid momentum and the growth of the recirculation zone. The total vortex boundary size along the walls (C) expanded from 1.81 units to 4.57 units.
- **Vortex Core Shifts:** The reattachment length measured from Point A extended from 3.63 units to 5.07 units at maximum velocity. Simultaneously, the vortex-core location shifted upward as the Reynolds number increased, with the vertical coordinate (y_v) increasing from approximately 0.38 to 0.74 units.
- **Solver Convergence:** The velocity components safely satisfied the 10^{-7} residual target threshold. The global continuity imbalance was reduced to approximately -9.24×10^{-16} , indicating excellent discrete mass conservation across the entire domain.

References

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Appendix

To maintain structural clarity and prevent visual clutter within the primary body of the report, detailed flow visualizations and validation curves are anchored to the baseline case ($Re = 60$, fine mesh tier). This appendix compiles the remaining computational solution verification plots and qualitative velocity streamline tracers extracted across the three mesh resolution tiers for the targeted Reynolds number configuration.

A: $Re = 60$

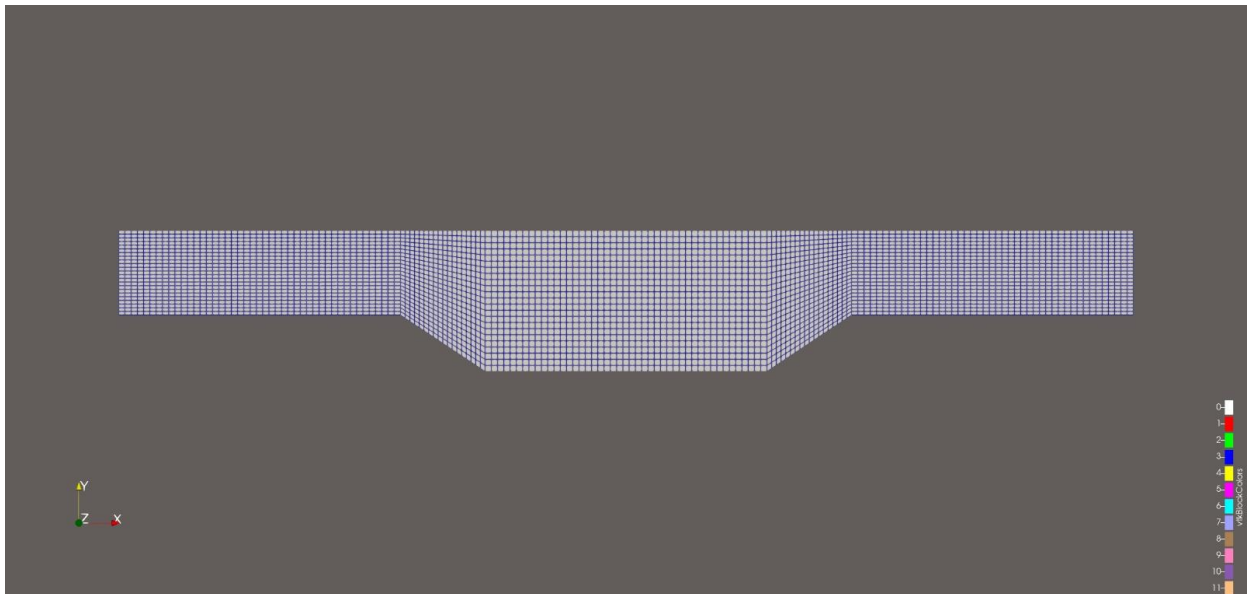


Fig A.1: Mesh of $Re = 60$ with coarse mesh

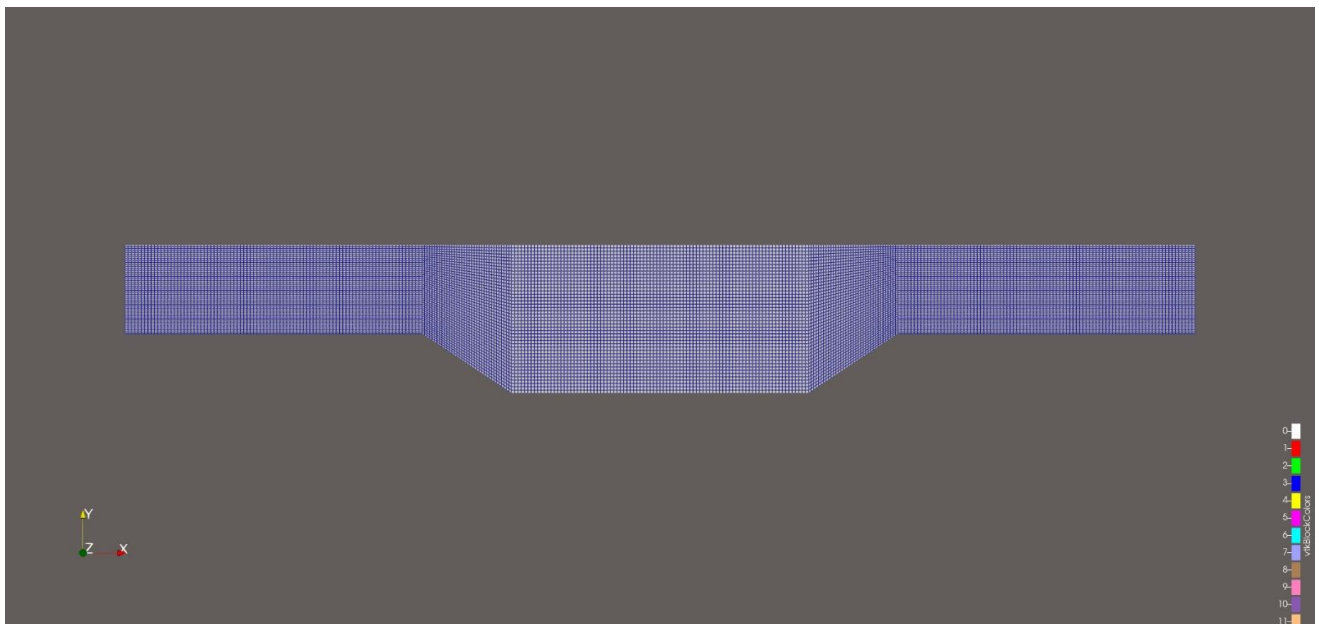


Fig A.2: Mesh of $Re = 60$ with medium mesh

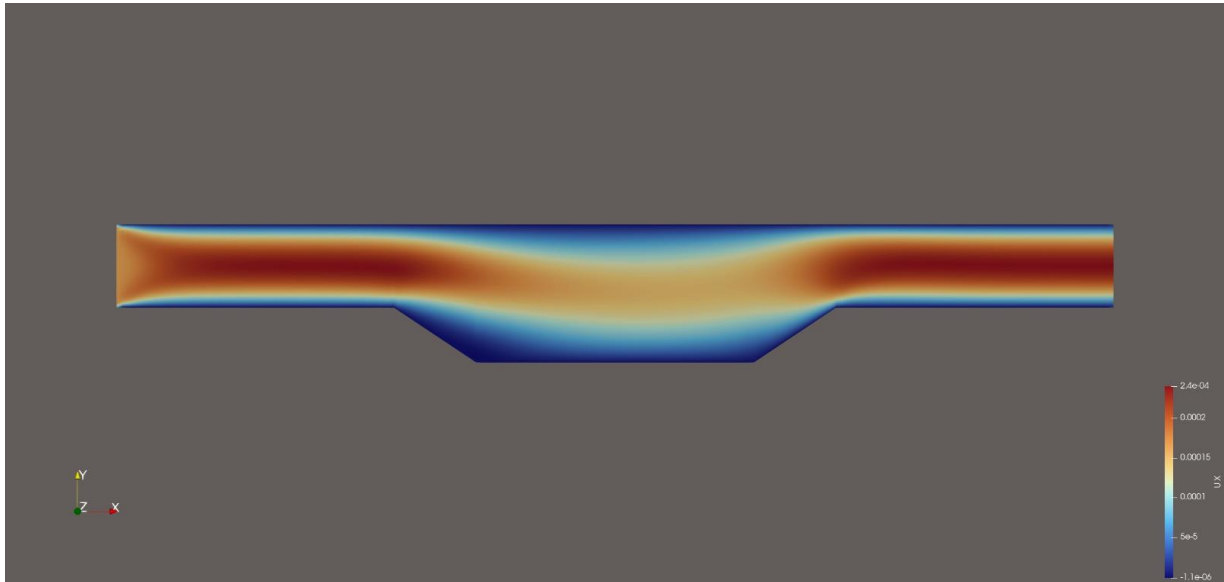


Fig A.3: Contour of Velocity along X direction of $Re = 60$ with coarse mesh

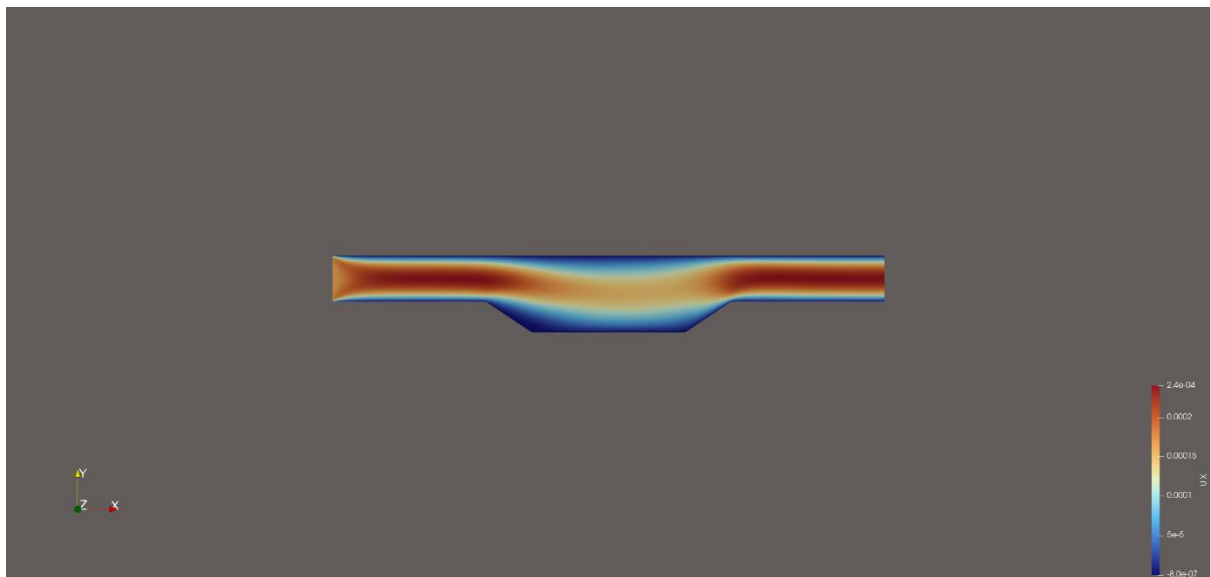


Fig A.4: Contour of Velocity along X direction of $Re = 60$ with medium mesh

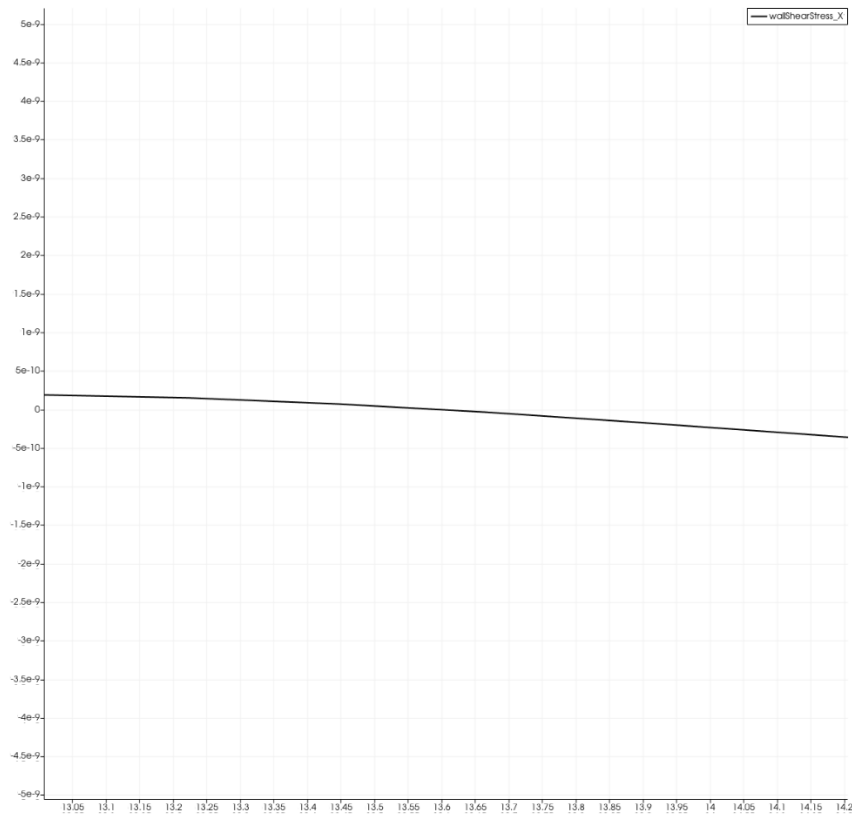


Fig A.5: wallShearStress_x of $Re = 60$ with coarse mesh along the horizontal floor from which reattachment length from Point A ($L_{r,A}$) and C_2 is calculated

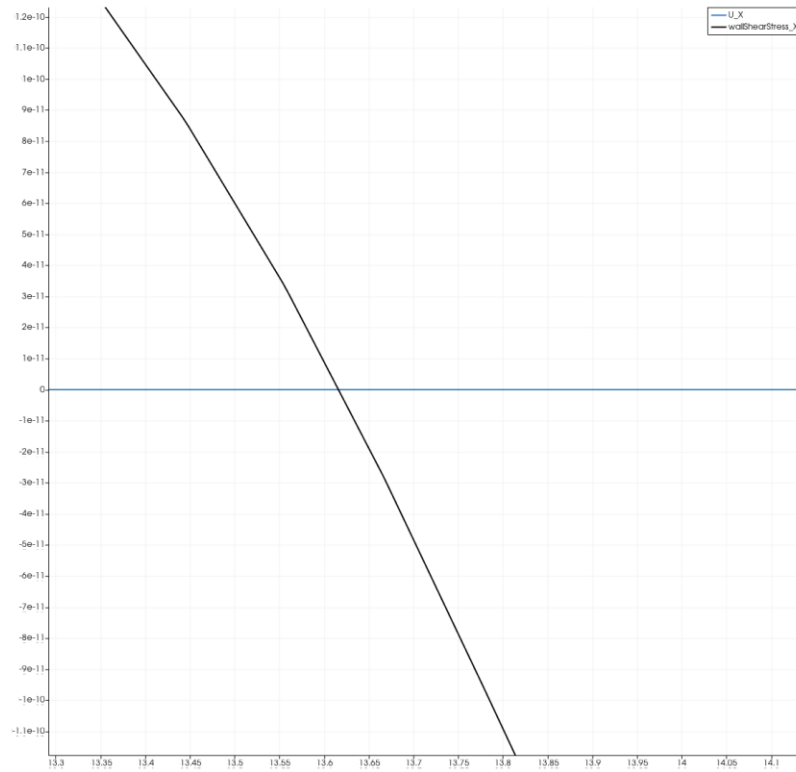


Fig A.6: $wallShearStress_x$ of $Re = 60$ with medium mesh along the horizontal floor from which reattachment length from Point A ($L_{r,A}$) and C_2 is calculated

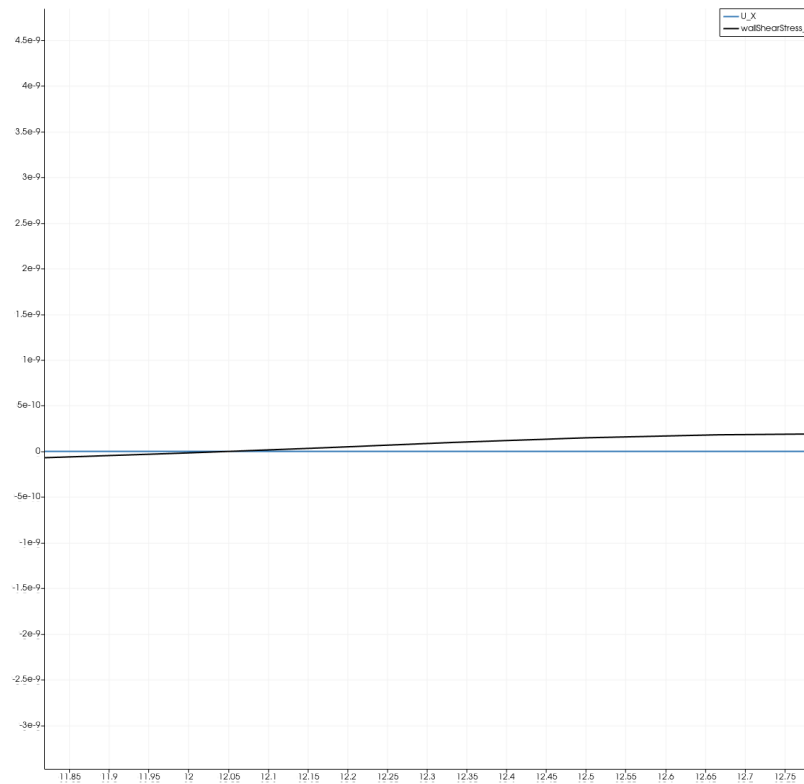


Fig A.7: wallShearStress_x of $Re = 60$ with coarse mesh along the slope from which C_1 is calculated

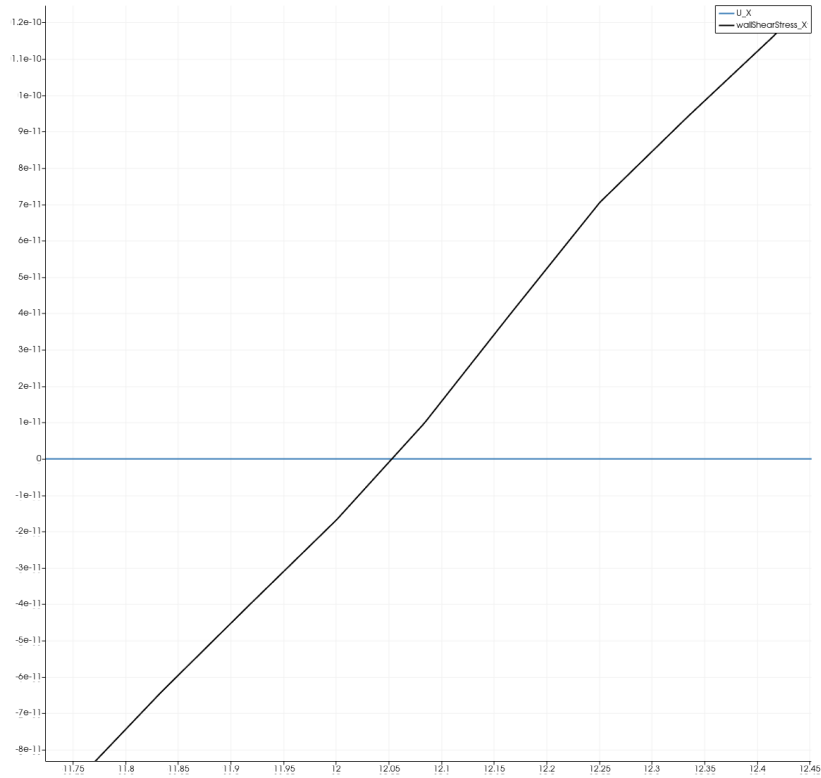


Fig A.8: wallShearStress_x of $Re = 60$ with medium mesh along the slope from which C_1 is calculated

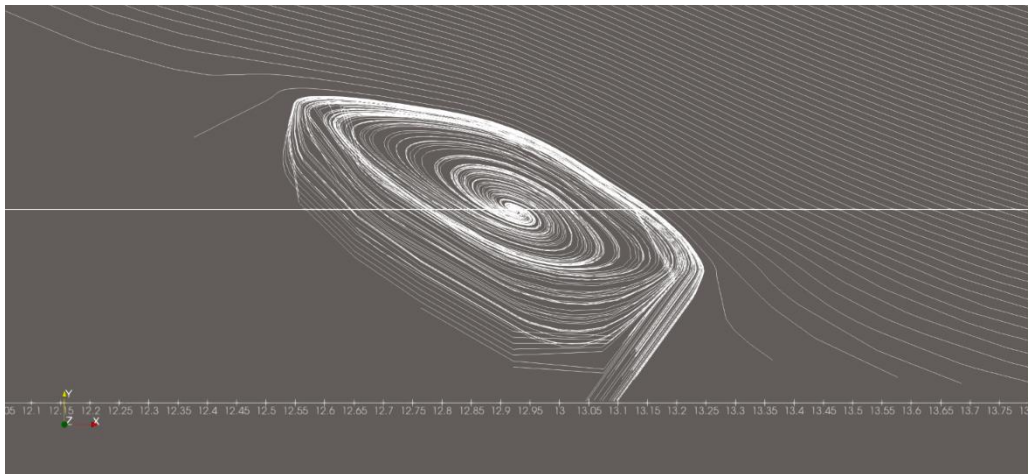


Fig A.9: Vortex formation of $Re = 60$ with coarse mesh depicted using Streamlines to approximate the centre of vortex visually

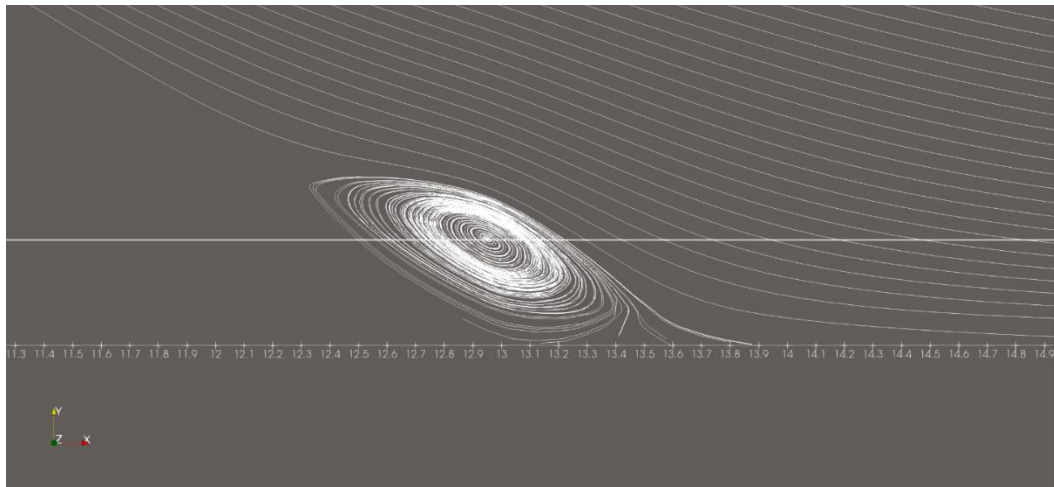


Fig A.10: Vortex formation of $Re = 60$ with medium mesh depicted using Streamlines to approximate the centre of vortex visually

B: $Re = 80$

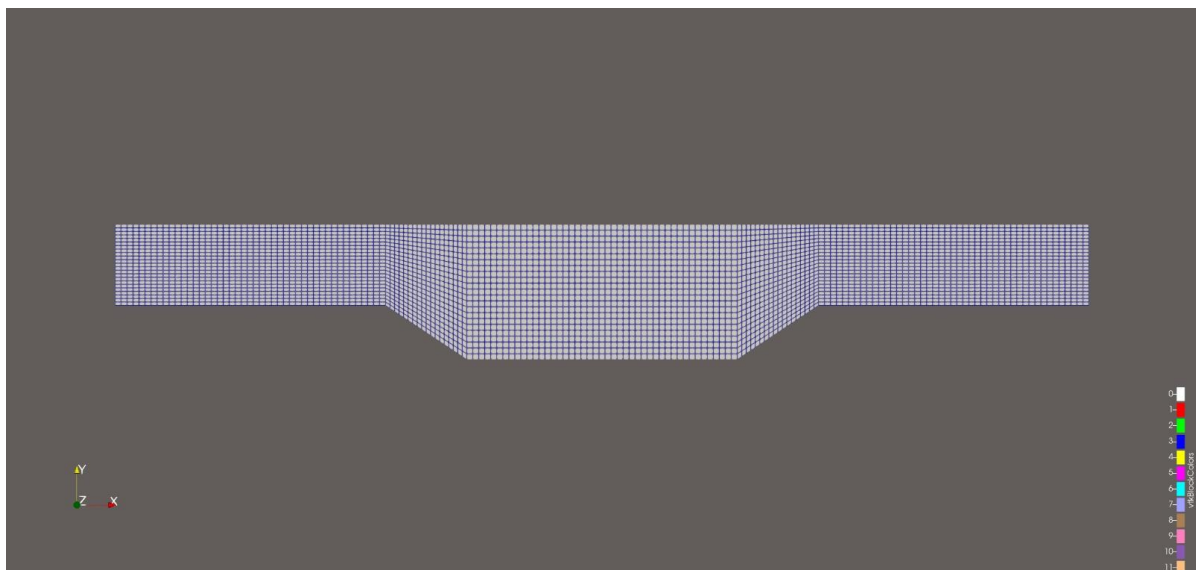


Fig B.1: Mesh of $Re = 80$ with coarse mesh

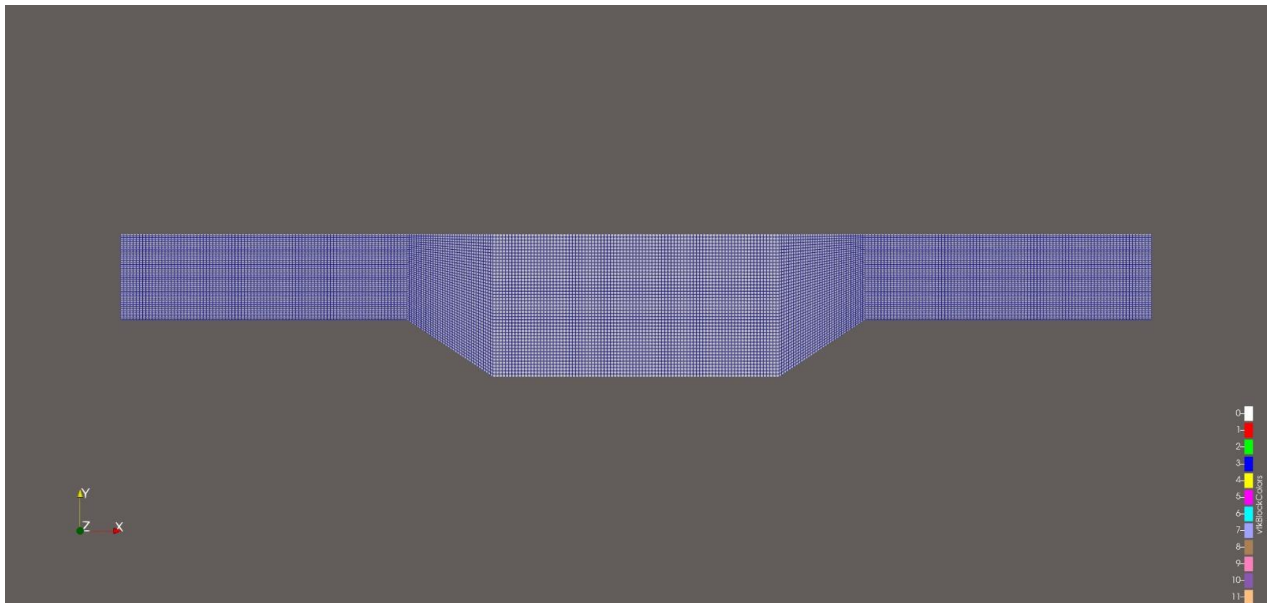


Fig B.2: Mesh of $Re = 80$ with medium mesh

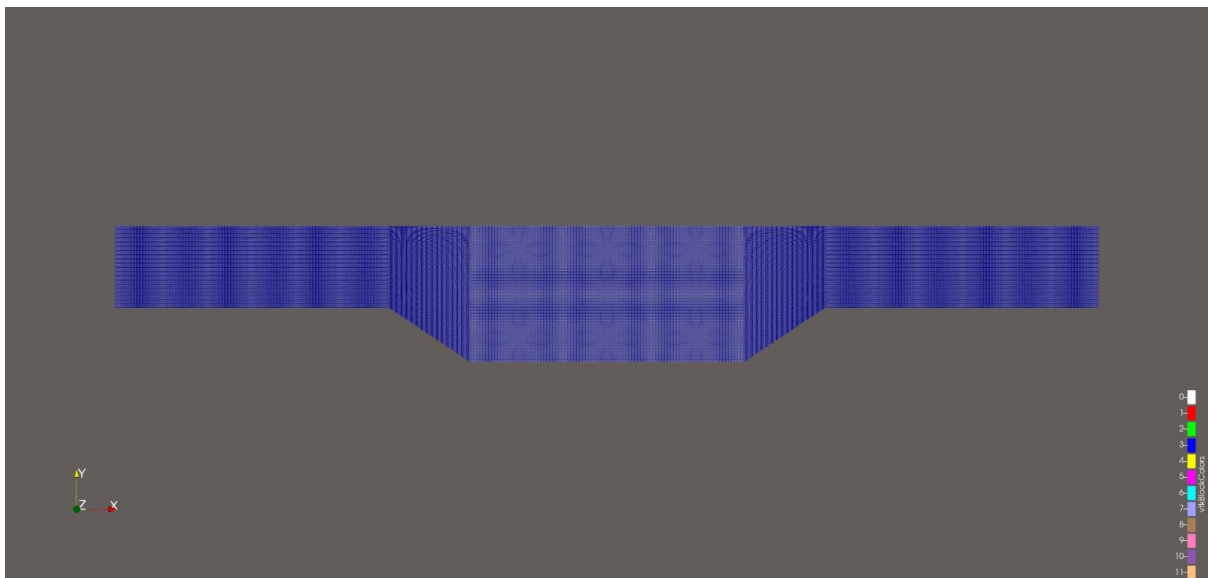


Fig B.3: Mesh of $Re = 80$ with fine mesh

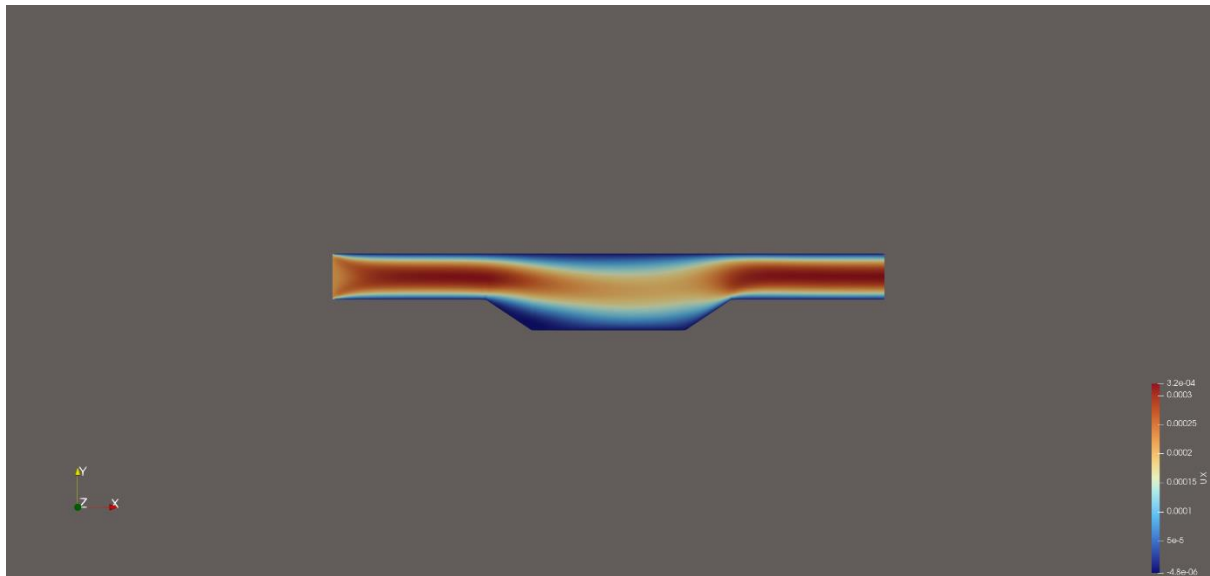


Fig B.4: Contour of Velocity along X direction of $Re = 80$ with coarse mesh

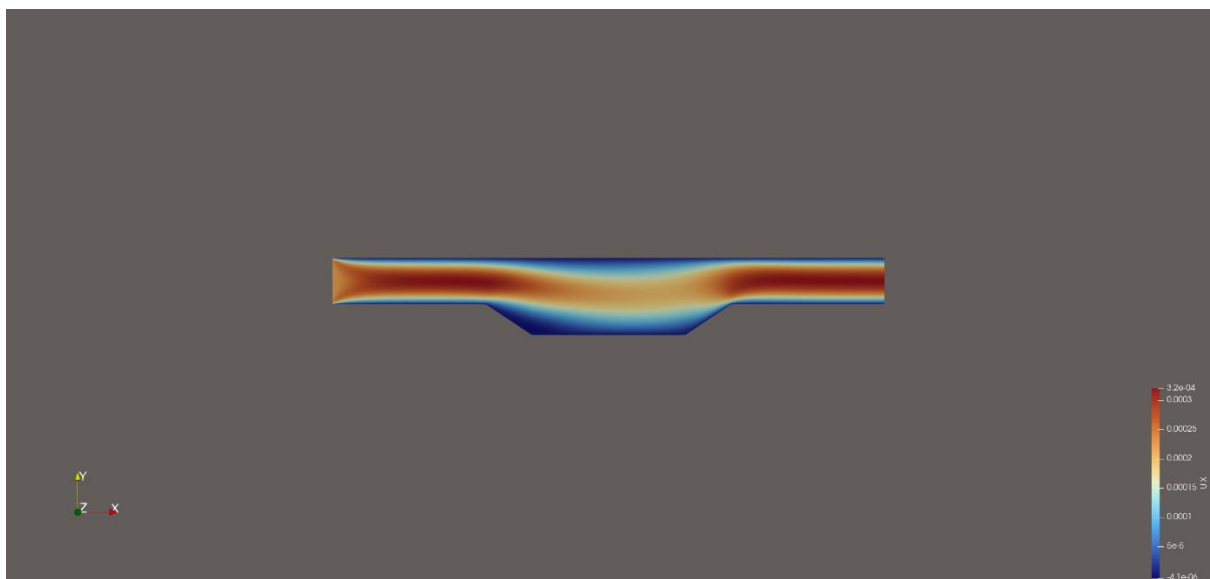


Fig B.5: Contour of Velocity along X direction of $Re = 80$ with medium mesh

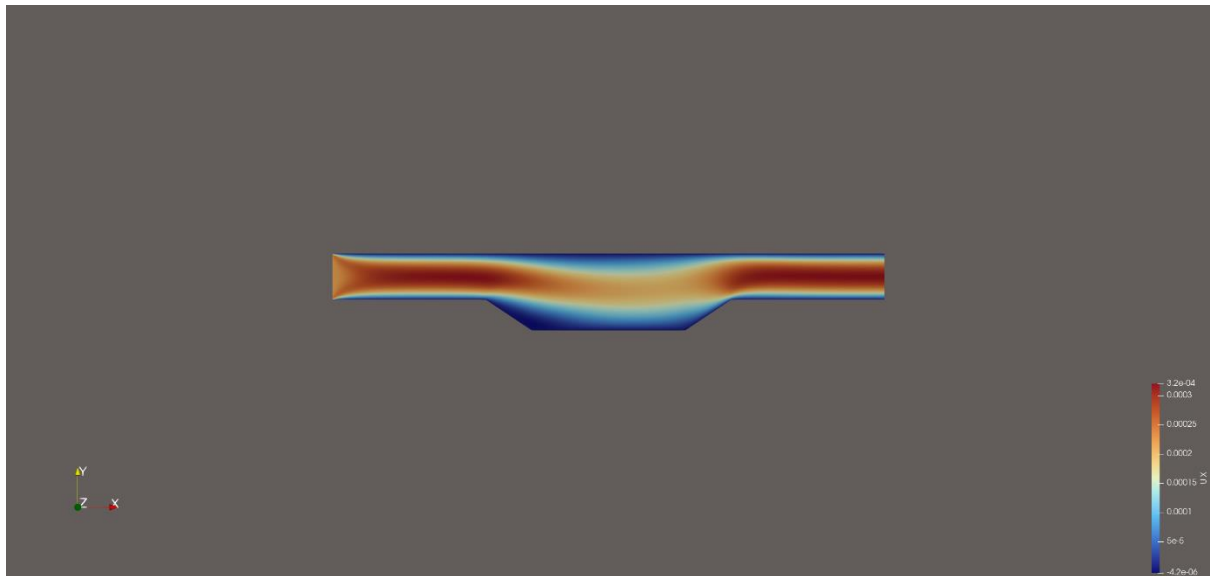


Fig B.6: Contour of Velocity along X direction of $Re = 80$ with fine mesh

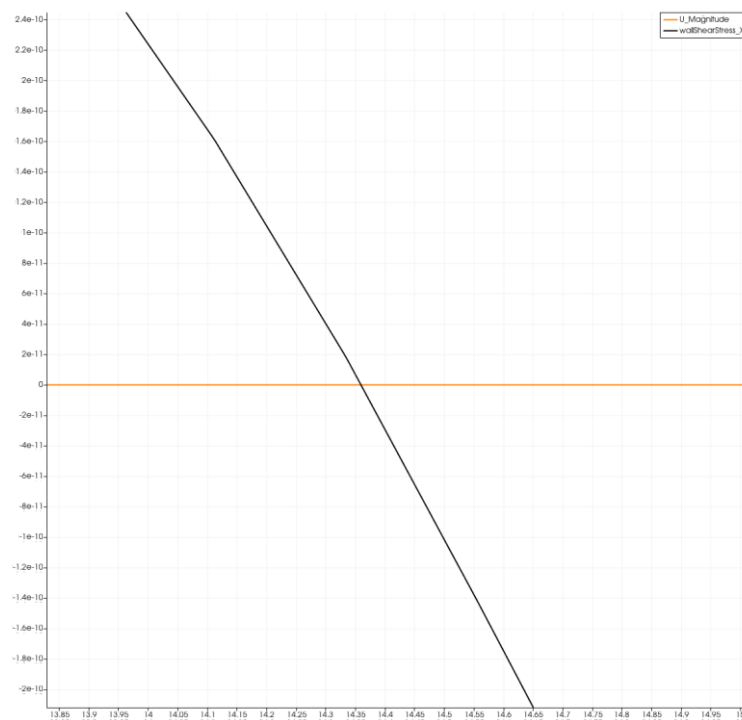


Fig B.7: wallShearStress_x of $Re = 80$ with coarse mesh along the horizontal floor from which reattachment length from Point A ($L_{r,A}$) and C_2 is calculated

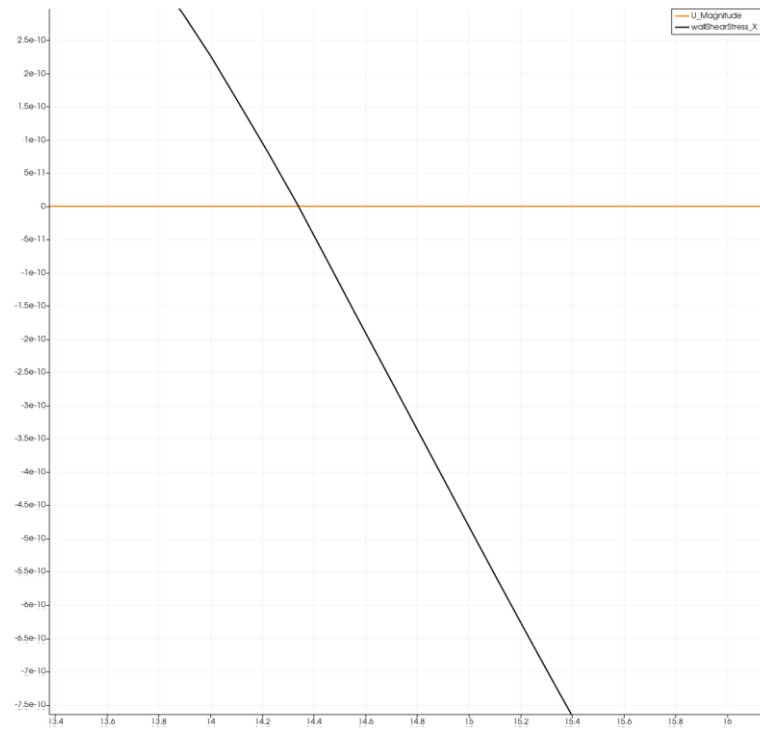


Fig B.8: wallShearStress_x of $Re = 80$ with medium mesh along the horizontal floor from which reattachment length from Point A ($L_{r,A}$) and C_2 is calculated

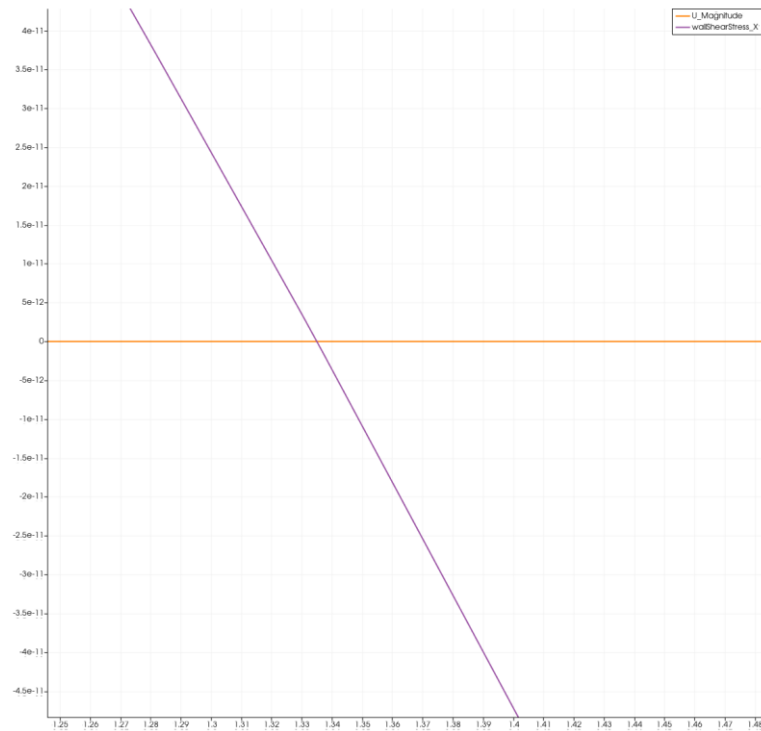


Fig B.9: wallShearStress_x of $Re = 80$ with fine mesh along the horizontal floor from which reattachment length from Point A ($L_{r,A}$) and C_2 is calculated

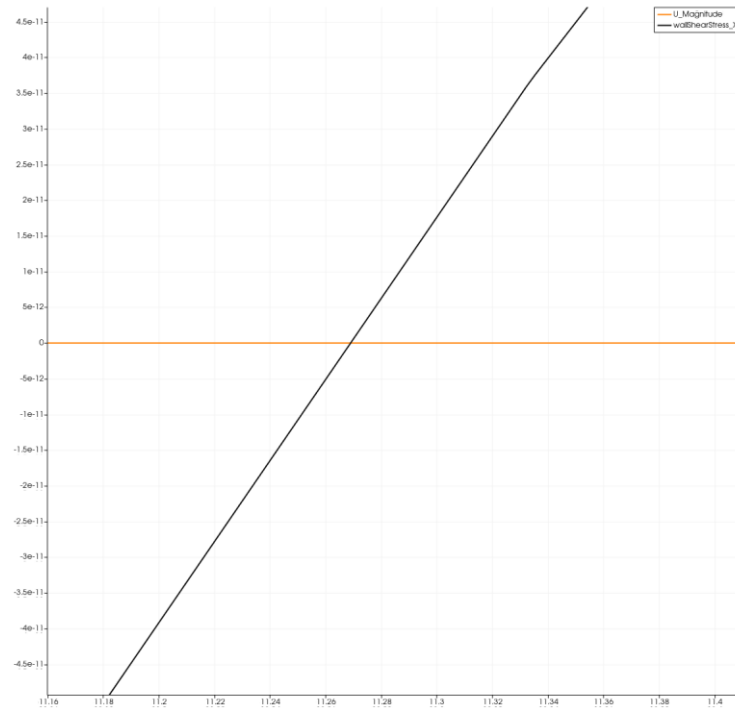


Fig B.10: wallShearStress_x of $Re = 80$ with coarse mesh along the slope from which C_1 is calculated

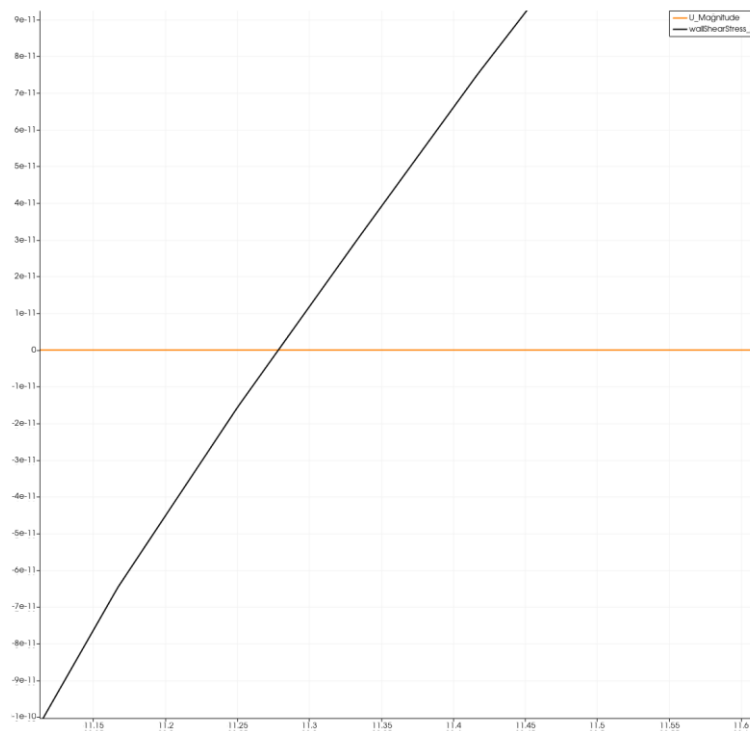


Fig B.11: wallShearStress_x of $Re = 80$ with medium mesh along the slope from which C_1 is calculated



Fig B.12: wallShearStress_x of $Re = 80$ with fine mesh along the slope from which C_1 is calculated

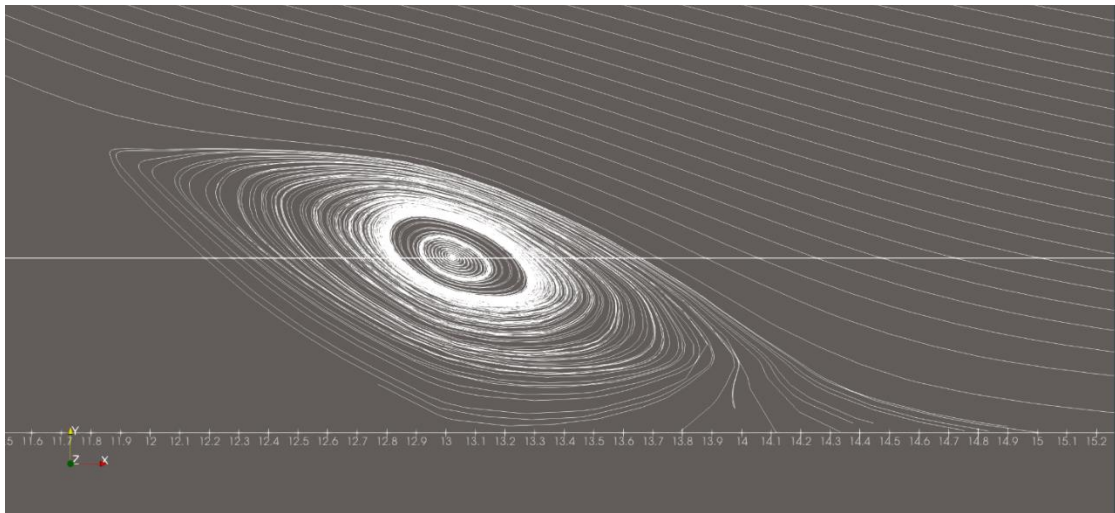


Fig B.13: Vortex formation of $Re = 80$ with coarse mesh depicted using Streamlines to approximate the centre of vortex visually

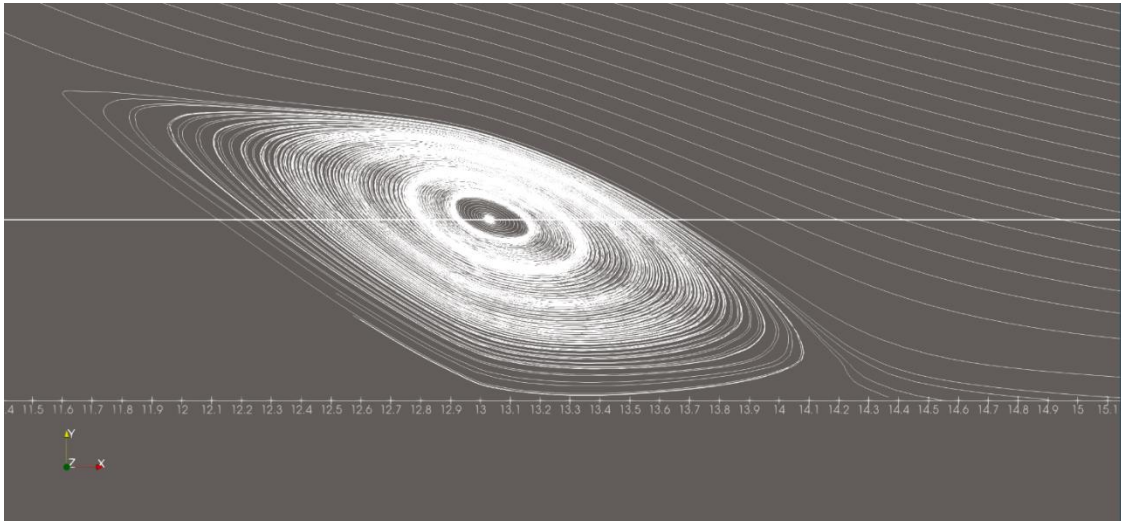


Fig B.14: Vortex formation of $Re = 80$ with medium mesh depicted using Streamlines to approximate the centre of vortex visually

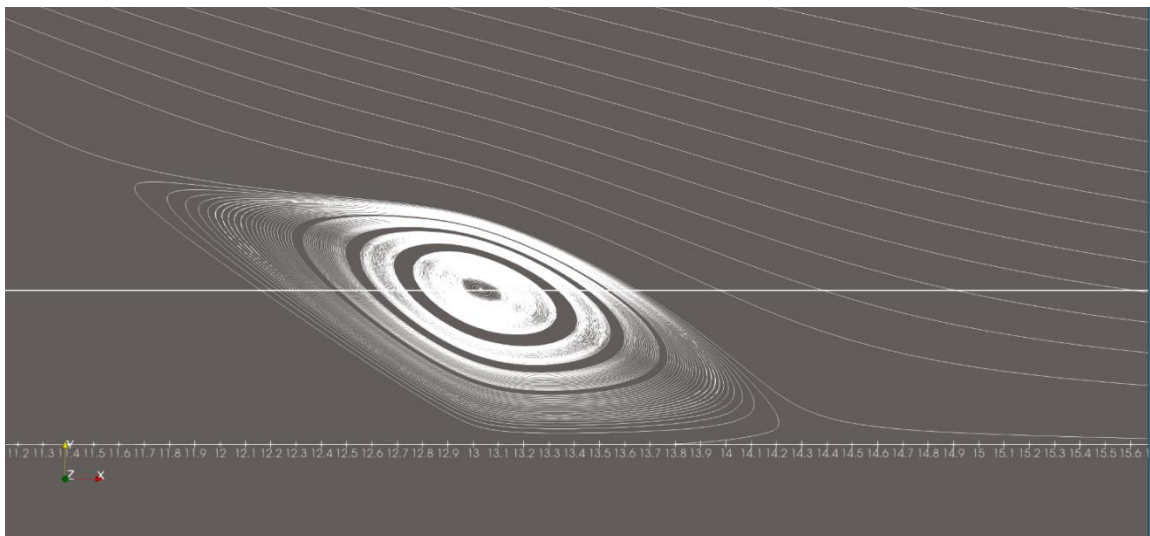


Fig B.15: Vortex formation of $Re = 80$ with fine mesh depicted using Streamlines to approximate the centre of vortex visually

C: $Re = 100$

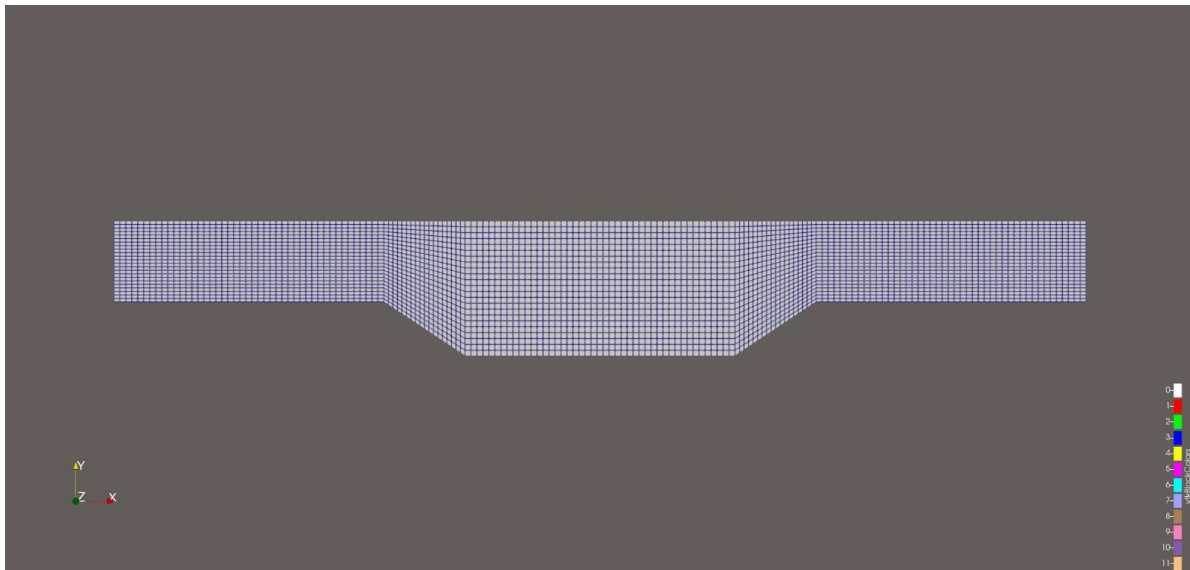


Fig C.1: Mesh of $Re = 100$ with coarse mesh

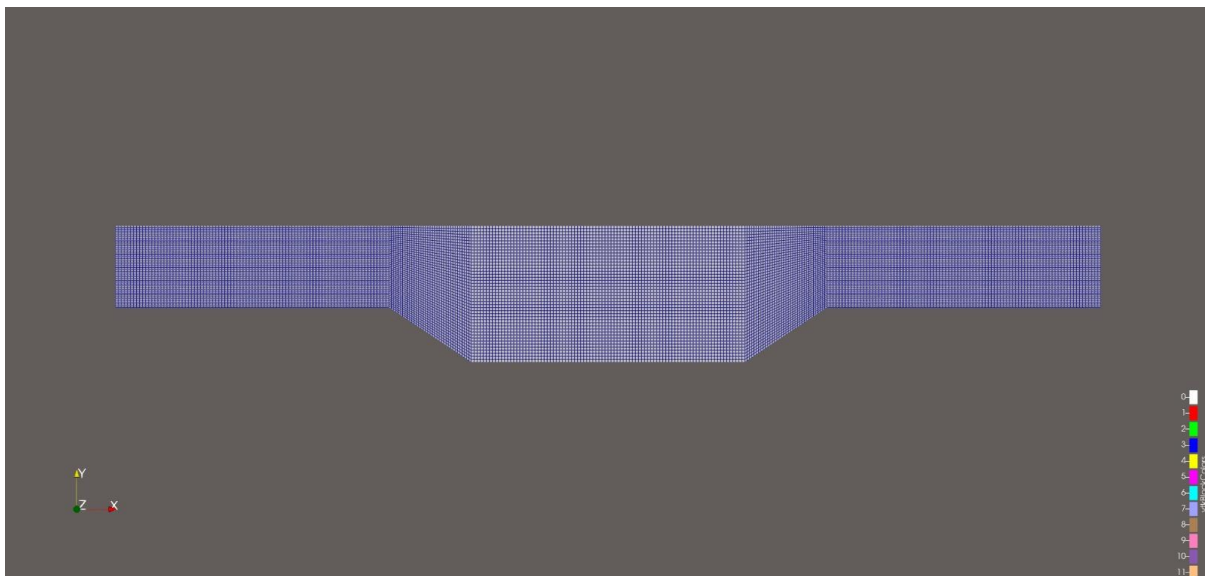


Fig C.2: Mesh of $Re = 100$ with medium mesh

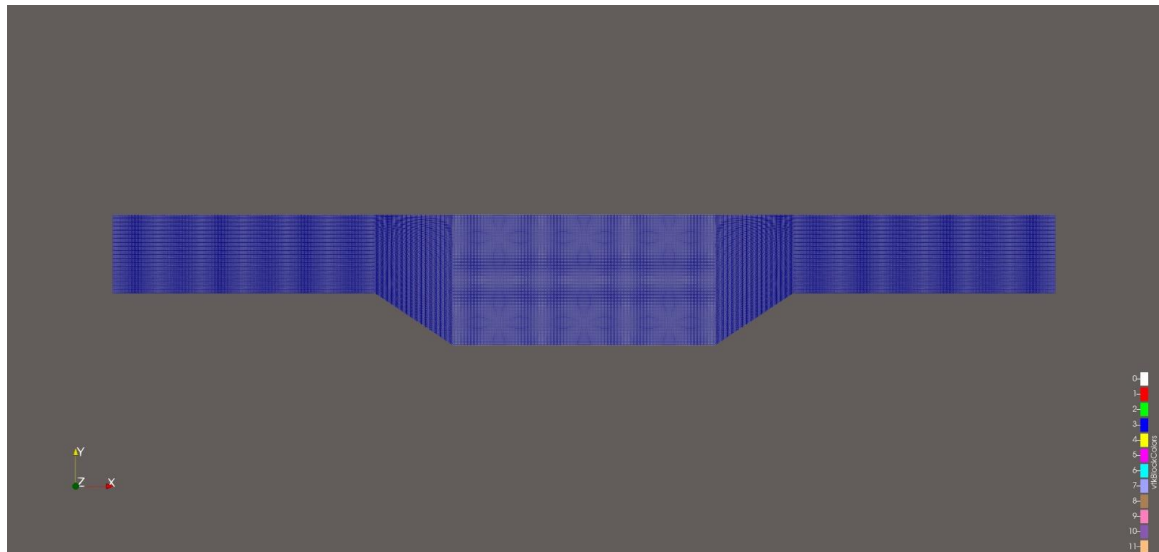


Fig C.3: Mesh of $Re = 100$ with fine mesh

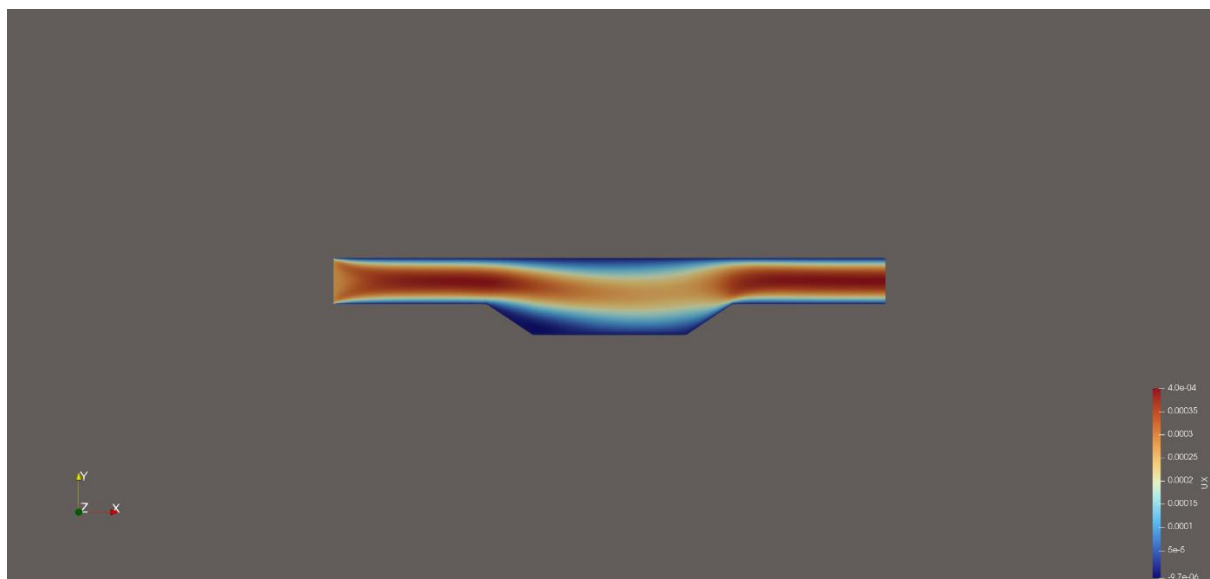


Fig C.4: Contour of Velocity along X direction of $Re = 100$ with coarse mesh

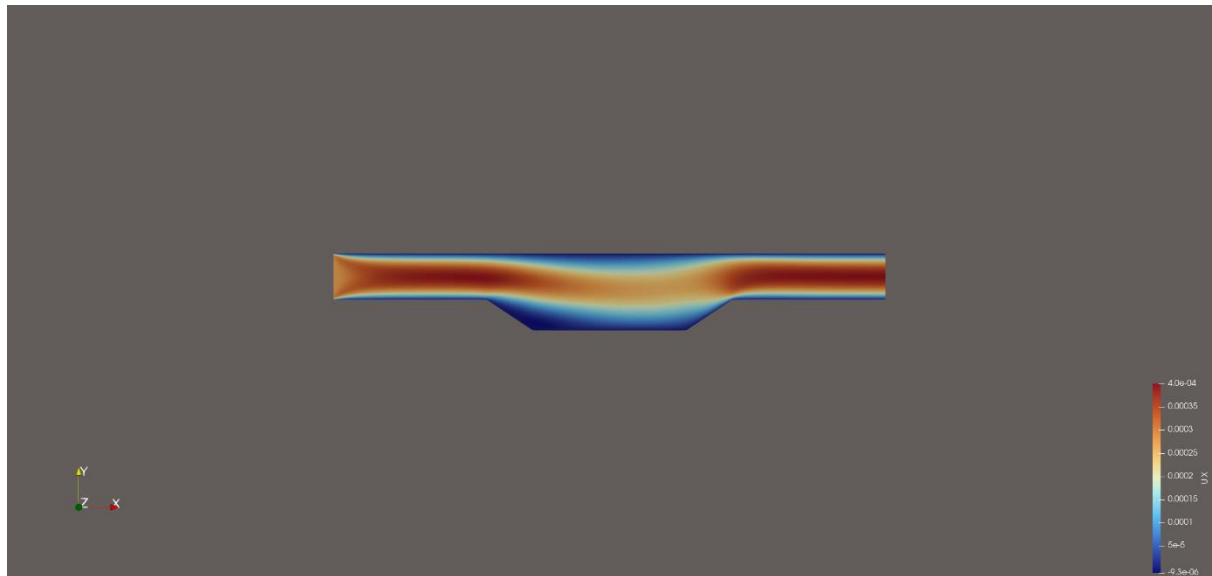


Fig C.5: Contour of Velocity along X direction of $Re = 100$ with medium mesh

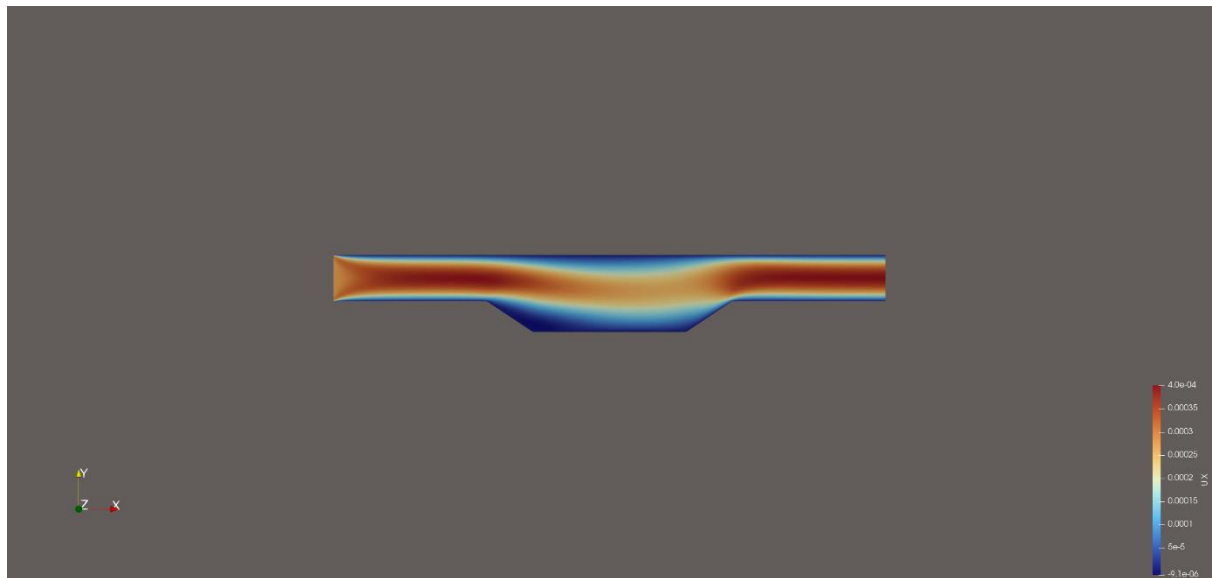


Fig C.6: Contour of Velocity along X direction of $Re = 100$ with fine mesh

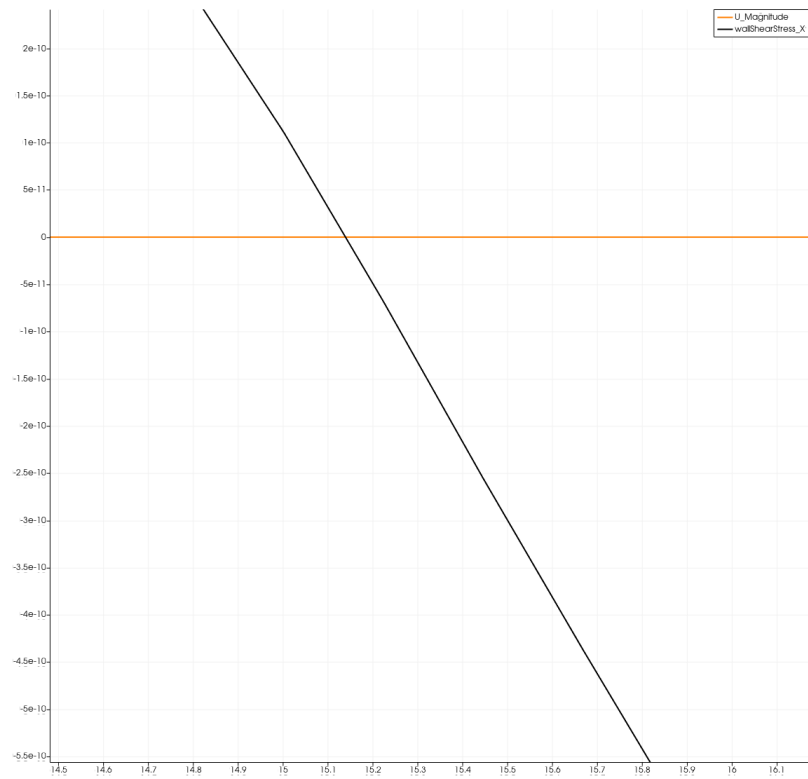


Fig C.7: wallShearStress_x of $Re = 100$ with coarse mesh along the horizontal floor from which reattachment length from Point A (L_r, A) and C2 is calculated

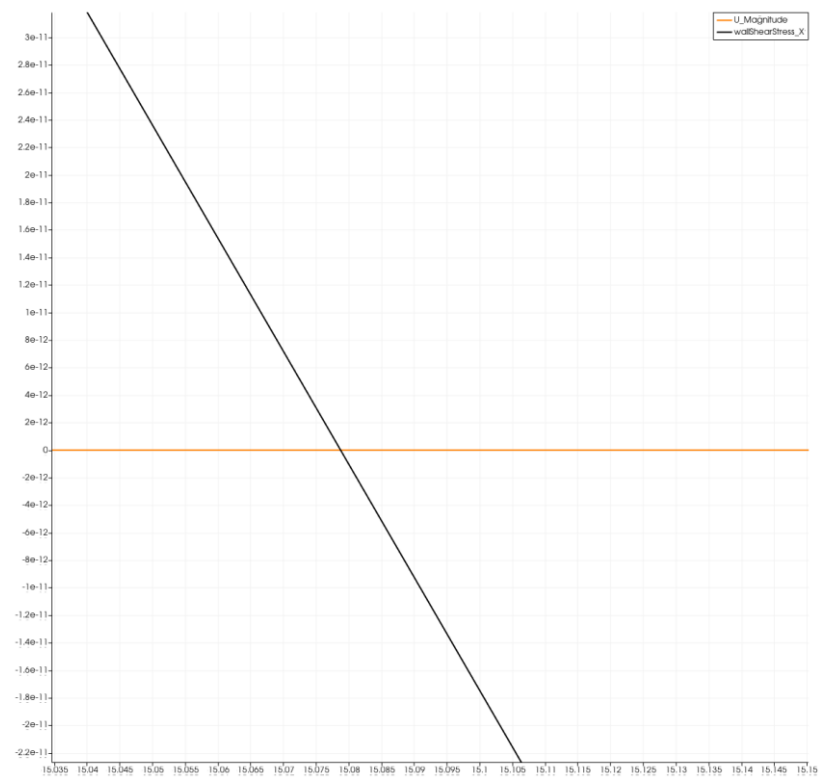


Fig C.8: wallShearStress_x of $Re = 100$ with medium mesh along the horizontal floor from which reattachment length from Point A (L_r, A) and C2 is calculated

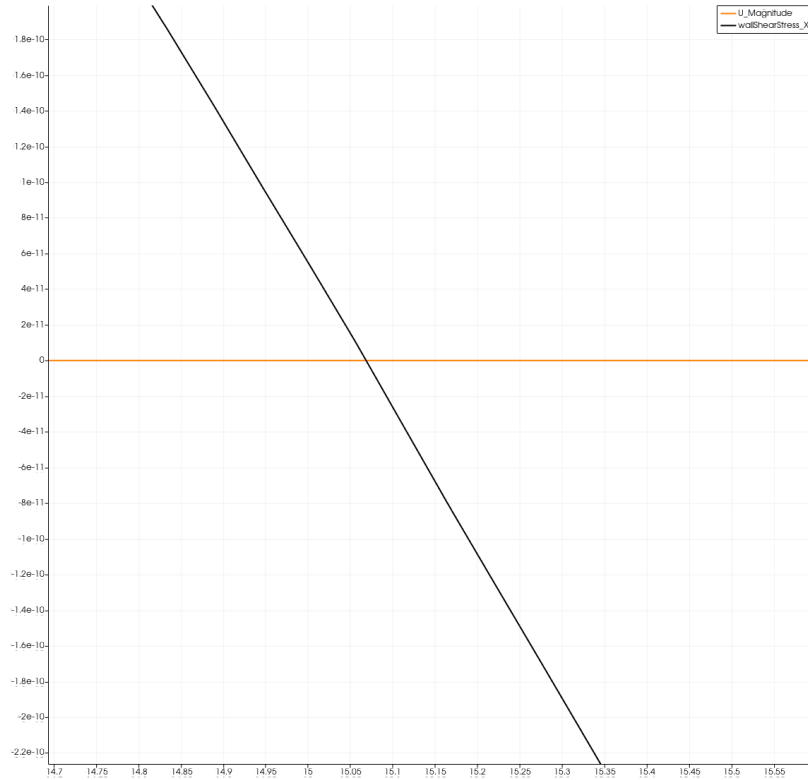


Fig C.9: wallShearStress_x of $Re = 100$ with fine mesh along the horizontal floor from which reattachment length from Point A (L_r, A) and C2 is calculated

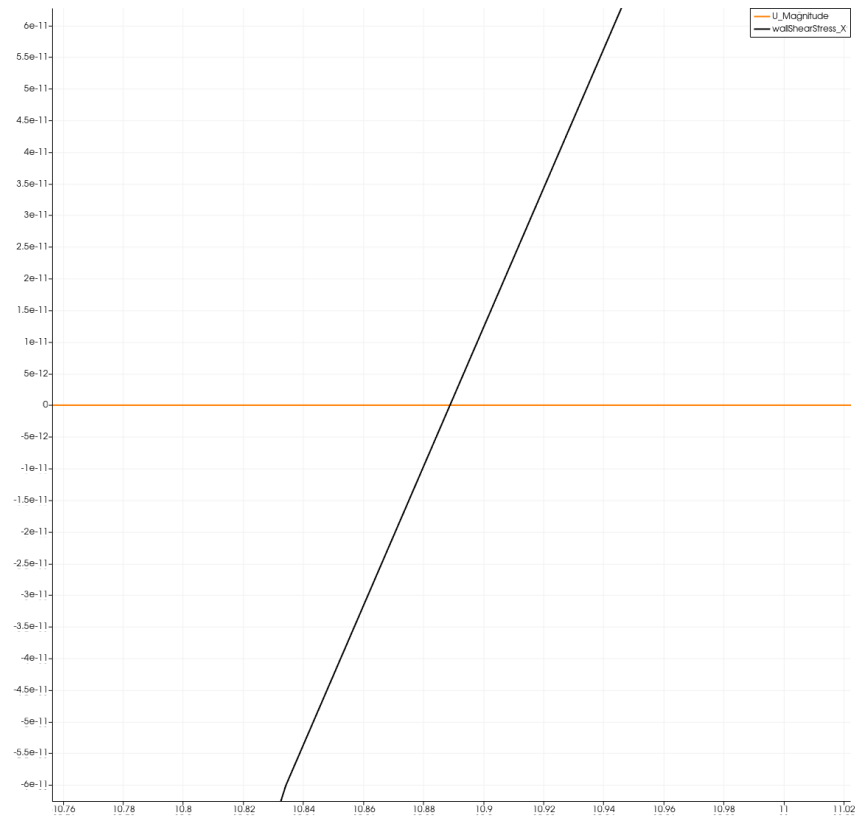


Fig C.10: wallShearStress_x of $Re = 100$ with coarse mesh along the slope from which C_1 is calculated

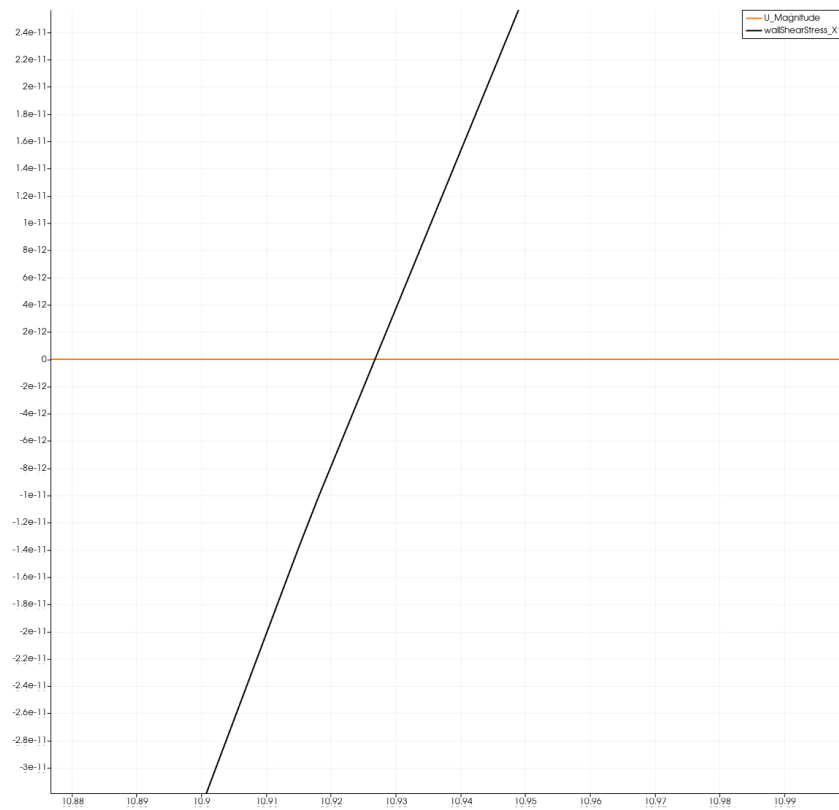


Fig C.11: wallShearStress_x of $Re = 100$ with medium mesh along the slope from which C_1 is calculated

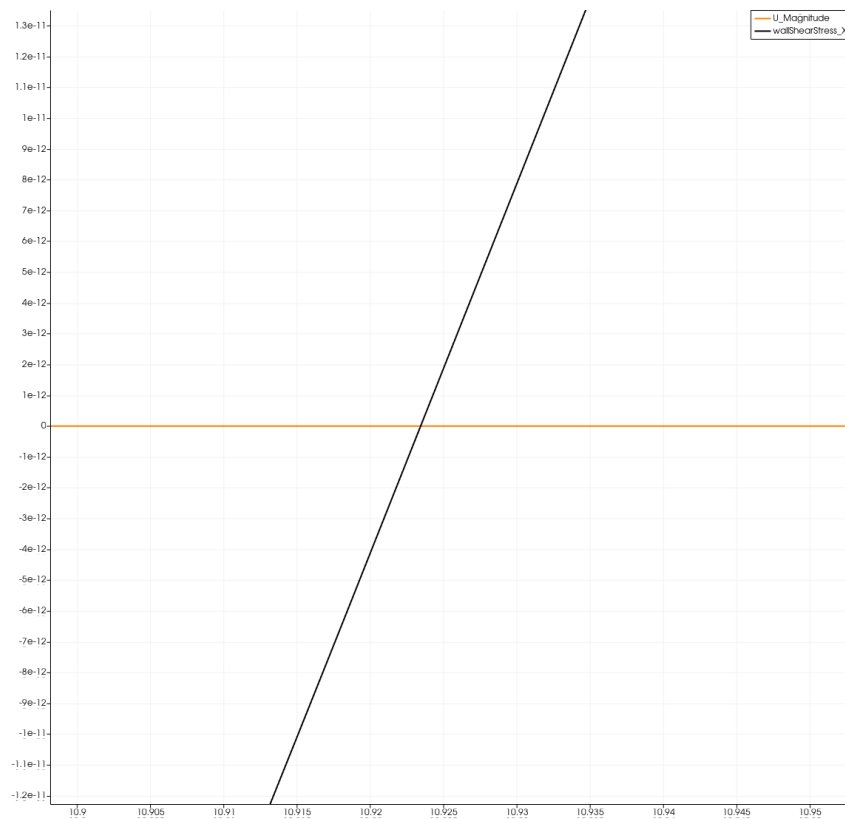


Fig C.12: wallShearStress_x of $Re = 100$ with fine mesh along the slope from which C_1 is calculated

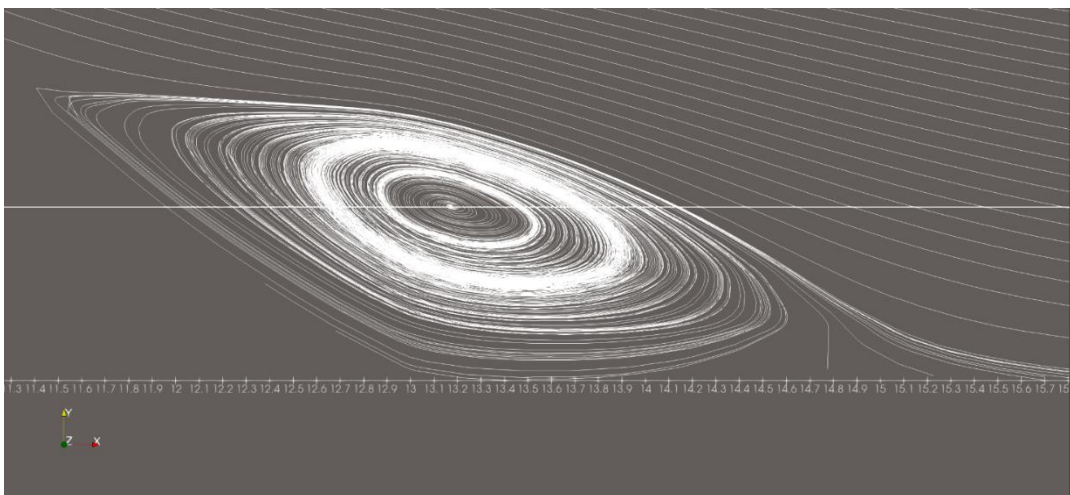


Fig C.13: Vortex formation of $Re = 100$ with coarse mesh depicted using Streamlines to approximate the centre of vortex visually

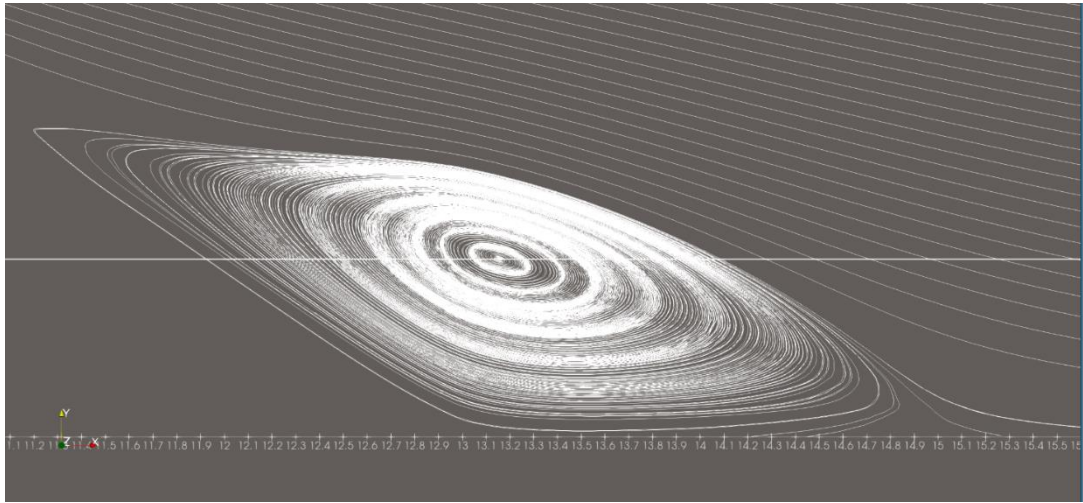


Fig C.14: Vortex formation of $Re = 100$ with medium mesh depicted using Streamlines to approximate the centre of vortex visually

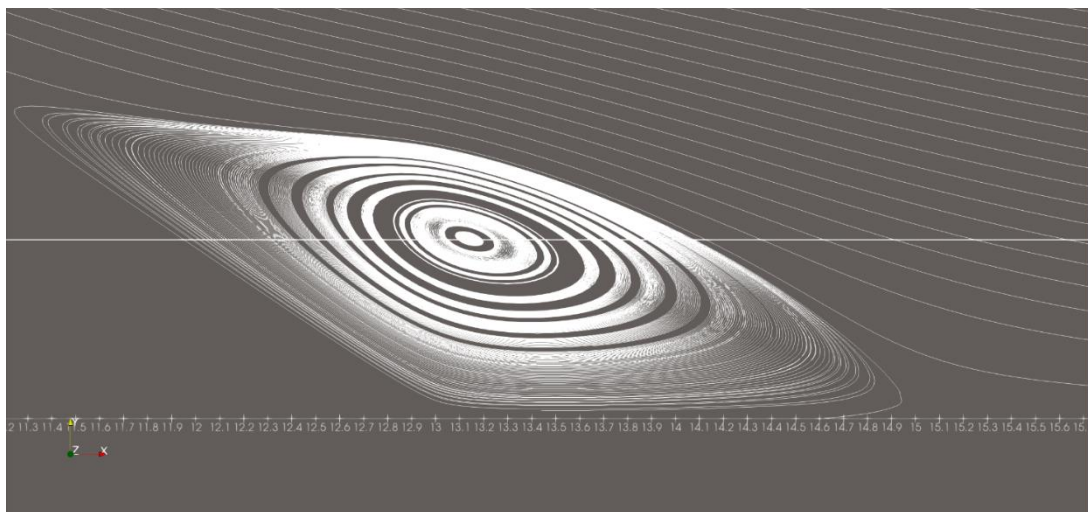


Fig C.15: Vortex formation of $Re = 100$ with fine mesh depicted using Streamlines to approximate the centre of vortex visually